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Abstract

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Chapter 1

A Bayesian Nonparametric Toolbox

1.1 Random Probability Measures and Exchangeability

In parametric Bayesian statistics, one works with statistical models, namely probability distributions indexed by finite-dimensional parameters. In many applications, this assumption can be restrictive. Bayesian nonparametric statistics therefore provides more flexible modeling approaches. The fundamental object to work with is a random probability measure, namely a random variable taking values in the space of probability measures that may have generated the observed data. Before formally defining this object, we introduce the setting for the statistical models.

Let $\mathbb{X}$ be a Polish space equipped with its Borel σ-algebra $\mathcal{B}(\mathbb{X})$. Denote by $\mathcal{P}(\mathbb{X})$ the space of probability measures on $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$. We equip $\mathcal{P}(\mathbb{X})$ with the Borel σ-algebra generated by the topology of weak convergence. This σ-algebra is the smallest one that makes the maps $\phi_A : \mathcal{P}(\mathbb{X}) \to [0, 1]$, defined by

$$\phi_A(P) = P(A),$$

measurable for all $A \in \mathcal{B}(\mathbb{X})$ (see Proposition A.5 in Ghosal and Van der Vaart (2017)).

Definition 1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Random probability measure is a measurable map from Ω to $\mathcal{P}(\mathbb{X})$.

In the literature, random probability measures are also often defined in terms of transition probability kernels. Under the measurable structure considered here, these two formulations are equivalent. We will denote the random probability measure by $\tilde{P}$.

A fundamental idea in Bayesian statistics is that, to make a statistical inference, the order in which the observations are observed should not matter. This symmetry is formalized by the notion of exchangeability, introduced by de Finetti.

Definition 1.2. A sequence of random variables $(X_i)_{i\geq 1}$ is exchangeable if

$$\big(X_1,\ldots,X_n\big) \overset{d}{=} \big(X_{\sigma(1)},\ldots,X_{\sigma(n)}\big),$$

for all $n \geq 1$ and every permutation σ of $\{1,\ldots,n\}$.

By de Finetti's representation theorem, exchangeability is equivalent to the assumption that the law of the observations is the mixture of product probability measures weighted by the prior distribution.

Theorem 1.3. *A sequence of random variables $(X_i)_{i\geq 1}$ is exchangeable if and only if there exists a probability measure $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$ such that*

$$\mathbb{P}\left(X_1 \in A_1,\ldots,X_n \in A_n\right) = \int_{\mathcal{P}(\mathbb{X})} \prod_{i=1}^{n} P(A_i)\,\mathrm{d}Q(P)$$

for all $n \geq 1$ and $A_1,\ldots,A_n \in \mathcal{B}(\mathbb{X})$.

A consequence of the above theorem is the following interpretation of the data generation process:

$$\begin{aligned}&\tilde{P} \sim Q\\ &X_1,\ldots,X_n|\tilde{P} \overset{\text{i.i.d}}{\sim} \tilde{P},\end{aligned}$$

where $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$ is referred as prior distribution.

In Bayesian nonparametrics, one of the early central challenges was to construct prior distributions on infinite-dimensional parameter spaces whose posterior distributions remained analytically or computationally tractable. A major breakthrough in this direction was the introduction of the Dirichlet process in Ferguson (1973). The Dirichlet process provides a prior distribution over random probability measures and possesses a conjugacy property analogous to that of finite-dimensional Bayesian models.

The method used by Ferguson to construct the Dirichlet process relies on finite-dimensional distributions. Specifically, one specifies the joint distribution of the probabilities assigned to finite measurable partitions and verifies the consistency conditions required to guarantee the existence of a random probability measure. This approach is analogous to the classical construction of stochastic processes through their finite-dimensional distributions. We now briefly describe this method.

Recall that, we aim to define a random variable $\tilde{P}$ taking values in the space of functions from $\mathcal{B}(\mathbb{X})$ to $[0,1]$, endowed with the additional properties required of a probability measure. The main tool for this construction is the Kolmogorov extension theorem.

This result allows one to define a stochastic process by specifying a collection of finite-dimensional distributions that satisfy suitable consistency conditions. In our setting, these finite-dimensional distributions are indexed by finite collections of measurable sets $A_1, \ldots, A_n \in \mathcal{B}(\mathbb{X})$. It is sufficient, however, to define these laws on measurable partitions of the space (see page 213 of Ferguson (1973)).

Definition 1.4. A measurable partition of a measurable space $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$ is a collection of sets $B_1, \ldots, B_k \in \mathcal{B}(\mathbb{X})$ such that the sets are pairwise disjoint and

$$\mathbb{X} = \bigcup_{i=1}^{k} B_i.$$

The first requirement is that the random measure assigns zero mass to the empty set, namely

$$\mathcal{L}\big(\tilde{P}(\emptyset)\big) = \delta_0.$$

The second requirement is the consistency condition. This condition states that if a measurable partition is refined into a finer partition, then the induced joint distributions must agree. More precisely, consider two measurable partitions $(B_1, \ldots, B_k)$ and $(C_1, \ldots, C_m)$, where each set B_i can be expressed as a union of consecutive elements of the finer partition: For all $i = 1, \ldots, k$,

$$B_i = \bigcup_{j=r_i}^{r_{i+1}-1} C_j,$$

with indices satisfying $1 = r_1 < r_2 < \cdots < r_{k+1} = m + 1$. Then the consistency condition requires that

$$\mathcal{L}\left(\sum_{j=r_1}^{r_2-1} \tilde{P}(C_j), \ldots, \sum_{j=r_k}^{r_{k+1}-1} \tilde{P}(C_j)\right) = \mathcal{L}\big(\tilde{P}(B_1), \ldots, \tilde{P}(B_k)\big).$$

Under these conditions, Lemma 1 of Ferguson (1973) guarantees the existence of a random probability measure whose finite-dimensional distributions coincide with those specified in the construction.

1.2 Completely Random Measures

Many Bayesian nonparametric models are constructed by first defining a random measure that need not be a probability measure and then transforming it into a probability measure. Completely random measures (CRM) provide a general framework for this construction.

Introduced in Kingman (1967), completely random measures can be viewed as measure-valued analogues of Lévy processes. In this setting, the index set is a σ-algebra of the underlying probability space, and the stochastic process takes the form of a random measure. Property of independent increments for Lévy processes is extended to this setting as independence of evaluations of a random measure on disjoint sets. In Bayesian nonparametric applications, such random measures are typically transformed, most commonly via normalization, into probability measures, which can then be used as priors over distributions. Let us start with the definition of completely random measures.

Although we assume throughout that $\mathbb{X}$ is a Polish space, in Kingman (1967) the space $\mathbb{X}$ is taken to be a general set whose σ-algebra contains all singletons. Since the Borel σ-algebra on a Polish space contains all singletons, this assumption is satisfied in our setting, and we may continue working with $\mathbb{X}$ Polish.

We denote by $\mathcal{M}_B(\mathbb{X})$ the space of finite, and by $\mathcal{M}(\mathbb{X})$ the space of locally finite measures. We endow $\mathcal{M}_B(\mathbb{X})$ with the topology of weak convergence. Since $\mathbb{X}$ is Polish, the space $\mathcal{M}_B(\mathbb{X})$ is also Polish. We equip $\mathcal{M}_B(\mathbb{X})$ with its Borel σ-algebra.

Definition 1.5. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A random measure $\tilde{\mu}$ is a map from Ω to $\mathcal{M}(\mathbb{X})$, such that, for all $A \in \mathcal{B}(\mathbb{X})$, the maps

$$\Omega \ni \omega \mapsto \tilde{\mu}(\omega)(A)$$

is measurable.

From now on, we will write $\tilde{\mu}(A)$ instead of $\tilde{\mu}(\omega)(A)$

Definition 1.6. A completely random measure $\tilde{\mu}$ is a random measure such that for all finite collection of mutually disjoint sets $A_1, \ldots, A_n \in \mathcal{B}(\mathbb{X})$, the random variables $\tilde{\mu}(A_1), \ldots, \tilde{\mu}(A_n)$ are independent.

1.2.1 Decomposition of Completely Random Measures

A key feature of completely random measures is that, despite their general definition, they admit a decomposition into simpler components. In particular, Kingman's representation theorem shows that a completely random measure can be decomposed into a deterministic part, a part supported on fixed atoms, and a purely random atomic part.

The main result in Kingman (1967) is the decomposition of a completely random measure. More precisely, let $\tilde{\mu}$ be a completely random measure satisfying the condition that there exists a countable collection $\{A_n\}$ of sets in $\mathcal{B}(\mathbb{X})$ such that $\mathbb{X} = \cup A_n$ and, for all $n \geq 1$,

$$\mathbb{P}\left(\tilde{\mu}(A_n) < \infty\right) > 0.$$

Then $\tilde{\mu}$ can be represented uniquely as

$$\tilde{\mu} = \mu + \sum_{x \in A} \tilde{\mu}\left(\{x\}\right) \delta_x + \sum_{i=1}^{\infty} J_i \delta_{X_i},$$

where μ is a deterministic non-atomic measure, A is a countable subset of $\mathbb{X}$, and $(J_i)_{i\geq 1}$ and $(X_i)_{i\geq 1}$ are independent sequences of random variables taking values in $(0,\infty)$ and $\mathbb{X}$, respectively.

In most of the literature, attention is restricted to the purely atomic random component of a CRM. This component can also be represented in terms of a Poisson random measure.

Recall that a Poisson random measure is a completely random measure and is characterized by its Lévy intensity ν, which is a σ-finite measure on $(0,\infty)\times\mathbb{X}$, and for every $C \in \mathcal{B}\left((0,\infty)\times\mathbb{X}\right)$ such that $\nu(C) < \infty$, one has

$$\mathcal{N}(C) \sim \mathrm{Poisson}(\nu(C)).$$

To avoid trivial cases, we assume that $\nu \neq 0$. Note that $\mathcal{N}$ is a discrete random measure. The purely atomic random component then admits the representation, for all $A \in \mathcal{B}(\mathbb{X})$,

$$\sum_{i=1}^{\infty} J_i \delta_{X_i}(A) = \int_{(0,\infty)\times A} s \, \mathrm{d}\mathcal{N}(s,x).$$

Moreover, the Lévy intensity ν associated with $\mathcal{N}$ satisfies the integrability condition

$$\int_{(0,\infty)\times\mathbb{X}} \min\{1,s\} \, \mathrm{d}\nu(s,x) < \infty.$$

Since, under this representation, the CRM is completely characterized by the Lévy intensity ν associated with the Poisson random measure $\mathcal{N}$, we write $\tilde{\mu} \sim \mathrm{CRM}(\nu)$. From now on, whenever we refer to CRMs, we mean the component admitting a representation in terms of a Poisson random measure. Note that, by defining the completely random measure as

$$\tilde{\mu}(A) = \int_{(0,\infty)\times A} s \, \mathrm{d}\mathcal{N}(s,x),$$

for $A \in \mathcal{B}(\mathbb{X})$, the fact that it takes values in $\mathcal{M}(\mathbb{X})$ and is measurable is guaranteed by the fact that $\mathcal{N}$ is a random measure.

1.2.2 Lévy Intensity and Laplace Functional

A completely random measure, similarly to a random probability measure as discussed in 1.1, can be uniquely identified by its finite-dimensional distributions, that is,

$$\mathcal{L}\left(\tilde{\mu}(A_1), \ldots, \tilde{\mu}(A_k)\right),$$

for a measurable partition $A_1, \ldots, A_k$ in $\mathcal{B}(\mathbb{X})$. By the independence assumption, these finite-dimensional distributions are identified once we specify the law

$$\mathcal{L}(\tilde{\mu}(A))$$

for $A \in \mathcal{B}(\mathbb{X})$. Equivalently, they are identified by the law of $\tilde{\mu}(f)$ for measurable $f : \mathbb{X} \to [0, \infty]$. For this reason, Laplace functionals play an important role.

For the CRM $\tilde{\mu}$, we have the characterization that, for all measurable functions $f : \mathbb{X} \to [0, \infty]$, or f integrable w.r.t $\tilde{\mu}$,

$$\mathbb{E}\left[e^{-\int_{\mathbb{X}} f(x) \mathrm{d}\tilde{\mu}(x)}\right] = \exp\left\{-\int_{\mathbb{R}_+ \times \mathbb{X}} \left(1 - e^{-sf(x)}\right) \mathrm{d}\nu(s, x)\right\}.$$

This result follows from Lemma J.2 in Ghosal and Van der Vaart (2017).

The integrability condition on the Lévy intensity has consequences for the completely random measure. In particular, for $\tilde{\mu} \sim \mathrm{CRM}(\nu)$, we have

$$\tilde{\mu}(\mathbb{X}) < \infty \quad \text{a.s.}$$

Indeed, note that

$$\tilde{\mu}(\mathbb{X}) = \int_{(0,1)\times\mathbb{X}} s \, \mathrm{d}\mathcal{N}(s, x) + \int_{(1,\infty)\times\mathbb{X}} s \, \mathrm{d}\mathcal{N}(s, x).$$

Let us first focus on the first term. By Campbell's theorem,

$$\mathbb{E}\left[\int_{(0,1)\times\mathbb{X}} s \, \mathrm{d}\mathcal{N}(s, x)\right] = \int_{(0,1)\times\mathbb{X}} s \, \mathrm{d}\nu(s, x) = \int_{(0,1)\times\mathbb{X}} \min\{s, 1\} \, \mathrm{d}\nu(s, x) < \infty.$$

Hence,

$$\int_{(0,1)\times\mathbb{X}} s \, \mathrm{d}\mathcal{N}(s, x) < \infty \quad \text{a.s.}$$

Next, we consider the second term. Since

$$\int_{(0,\infty)\times\mathbb{X}} \min\{1, s\} \, \mathrm{d}\nu(s, x) < \infty,$$

it follows that

$$\nu\big((1,\infty)\times\mathbb{X}\big)<\infty.$$

Therefore, since $\mathcal{N}\big((1,\infty)\times\mathbb{X}\big)\sim\text{Poisson}\big(\nu((1,\infty)\times\mathbb{X})\big)$, we have that $\mathcal{N}$ is a finite discrete measure. Consequently,

$$\int_{(1,\infty)\times\mathbb{X}} s\,\mathrm{d}\mathcal{N}(s,x)<\infty \quad \text{a.s.}$$

It is also important to note that a sufficient condition for $\tilde{\mu}(\mathbb{X})>0$ almost surely is

$$\nu((0,1)\times\mathbb{X})=\infty.$$

Indeed, since ν is σ-finite, we can find a sequence of sets $(B_i)_{i\geq 1}$ such that

$$B_i\subset B_{i+1},\quad \nu(B_i)<\infty,\quad \bigcup_{i\geq 1}B_i=(0,\infty)\times\mathbb{X}.$$

Then

$$\mathbb{P}(\mathcal{N}(B_i)=0)=e^{-\nu(B_i)}.$$

Moreover, it follows that

$$\mathbb{P}\left(\mathcal{N}((0,\infty)\times\mathbb{X})=0\right)\leq\inf_{i\geq 1}\mathbb{P}\left(\mathcal{N}(B_i)=0\right)=\inf_{i\geq 1}e^{-\nu(B_i)}=0.$$

Thus,

$$\mathcal{N}((0,\infty)\times\mathbb{X})>0 \quad \text{a.s.},$$

and we conclude that

$$\tilde{\mu}(\mathbb{X})>0 \quad \text{a.s.}$$

Since a Poisson random measure is a discrete random measure of the form

$$\sum_{i\geq 1}\delta_{(J_i,X_i)},$$

the random variables J_i and X_i are called jumps and atoms, respectively.

The Lévy intensity governs the law of the jumps and atoms. By the disintegration theorem, it can be decomposed into a component corresponding to the distribution of atom locations and another component describing the distribution of jump sizes at each location.

In particular, let us denote by $\mathcal{M}_1(\nu)$ the first moment of ν, defined as

$$\mathcal{M}_1(\nu) := \int_{(0,\infty)\times\mathbb{X}} s \, \mathrm{d}\nu(s,x) < \infty.$$

Then, for $\tilde{\mu} \sim \mathrm{CRM}(\nu)$ with finite mean, it follows from Proposition 3 in Catalano and Lavenant (2023) that

$$\mathrm{d}\nu(s,x) = \mathrm{d}\rho_x(s) \, \mathrm{d}P_0(x),$$

where P_0 is a probability measure on $\mathbb{X}$ and $\rho : \mathbb{X} \times \mathcal{B}((0,\infty)) \to [0,\infty]$ is a transition kernel. Whenever ρ does not depend on x, such a CRM is referred to as a homogeneous CRM.

1.2.3 Normalization and Identifiability

Completely random measures can be transformed into random probability measures. The transformation we consider in this chapter is normalization. In particular, for $\tilde{\mu} \sim \mathrm{CRM}(\nu)$, we consider

$$\tilde{\mu} \mapsto \frac{\tilde{\mu}}{\tilde{\mu}(\mathbb{X})}.$$

As noted in the previous subsection, to guarantee that this transformation is well defined, we need the Lévy intensity to satisfy an integrability condition and $\nu((0,\infty)\times\mathbb{X}) = \infty$. However, for the identification purposes that we will explain below, together with the integrability condition, we will assume a stronger condition, namely that

$$\rho_x((0,\infty)) = \infty \quad \text{for } P_0\text{-almost every } x \in \mathbb{X},$$

where ρ corresponds to the density of jumps in the decomposition of the Lévy intensity. CRMs satisfying this condition are referred to as infinitely active CRMs.

A crucial point to note in normalization is that there may exist two CRMs $\tilde{\mu}^1, \tilde{\mu}^2$ such that

$$\tilde{\mu}^1 \overset{d}{\neq} \tilde{\mu}^2, \qquad \text{but} \qquad \frac{\tilde{\mu}^1}{\tilde{\mu}^1(\mathbb{X})} \overset{d}{=} \frac{\tilde{\mu}^2}{\tilde{\mu}^2(\mathbb{X})}.$$

Therefore, if we work with a distance defined at the level of CRMs, we encounter an identifiability issue: two distinct CRMs may induce the same random probability measure after normalization, yet have strictly positive distance.

To overcome this issue, Catalano and Lavenant (2023) provides conditions under which two CRMs induce the same random probability measure. This allows one to work with a representative element from each equivalence class of CRMs that yield the same normalized measures.

The representative element within each equivalence class of CRMs that induce the same random probability measure through normalization is given by the scaled CRM, defined as follows.

Definition 1.7. A completely random measure $\tilde{\mu}$ is called a scaled CRM if

$$\mathbb{E}[\tilde{\mu}(\mathbb{X})] = 1.$$

If $\tilde{\mu}$ is a CRM with finite mean, meaning that $\mathcal{M}_1(\nu) < \infty$, its associated scaled CRM is defined by

$$\tilde{\mu}_S = \frac{\tilde{\mu}}{\mathbb{E}[\tilde{\mu}(\mathbb{X})]}.$$

Then, for two infinitely active completely random measures $\tilde{\mu}^1, \tilde{\mu}^2$ with finite means and associated base measures P_0^1, P_0^2 that are non-Dirac, Corollary 8 in Catalano and Lavenant (2023) states that

$$\frac{\tilde{\mu}^1}{\tilde{\mu}^1(\mathbb{X})} \overset{d}{=} \frac{\tilde{\mu}^2}{\tilde{\mu}^2(\mathbb{X})} \quad \Longleftrightarrow \quad \tilde{\mu}_S^1 \overset{d}{=} \tilde{\mu}_S^2.$$

Note that scaled CRMs are still completely random measures. Lemma 9 in Catalano and Lavenant (2023) provides a way to characterize the Lévy intensity of the scaled CRM.

In particular, let $\tilde{\mu} \sim \mathrm{CRM}(\nu)$ be such that

$$\mathrm{d}\nu(s,x) = \rho_x(s)\,\mathrm{d}s\,\mathrm{d}P_0(x).$$

Then the Lévy intensity ν_S of the scaled CRM $\tilde{\mu}_S$ is given by

$$\mathrm{d}\nu_S(s,x) = \mathbb{E}[\tilde{\mu}(\mathbb{X})]\,\rho_x\big(\mathbb{E}[\tilde{\mu}(\mathbb{X})]\,s\big)\,\mathrm{d}s\,\mathrm{d}P_0(x).$$

1.2.4 Posterior for Completely Random Measures

One of the main reasons for working with completely random measures is their flexibility for modelling and the tractability of the corresponding posterior distributions. We now describe the general form of the posterior distribution for CRMs, as developed in James et al. (2009). Consider the Bayesian model

$$\begin{aligned} \tilde{\mu} &\sim \mathrm{CRM}(\nu), \\ X_1, \ldots, X_n \mid \tilde{\mu} &\overset{\text{i.i.d}}{\sim} \frac{\tilde{\mu}}{\tilde{\mu}(\mathbb{X})}. \end{aligned} \tag{1.1}$$

The prior distribution is updated through the posterior distribution. Formally, the posterior is a transition kernel, denoted by $\mathcal{L}(\tilde{\mu} \mid X_{1:n})$, which is a mapping

$$\mathcal{L}(\tilde{\mu} \mid X_{1:n}) : \mathbb{X}^n \times \mathcal{B}(\mathcal{M}_B(\mathbb{X})) \to [0, 1],$$

where $\mathcal{B}(\mathcal{M}_B(\mathbb{X}))$ denotes the Borel σ-algebra on the space of finite measures on $\mathbb{X}$. For a fixed sequence of observations $x_1, \ldots, x_n$, we denote by

$$\tilde{\mu}^* \sim \mathcal{L}(\tilde{\mu} \mid X_{1:n} = x_{1:n})$$

a posterior random measure.

Before stating the main theorem, we introduce some additional notation. Given observations $x_1, \ldots, x_n$, let $x_1^*, \ldots, x_k^*$ denote the $k \leq n$ distinct values, and let

$$n_i := \#\{j : x_j = x_i^*\}, \qquad i = 1, \ldots, k,$$

be their corresponding multiplicities.

We also introduce an auxiliary random variable U with density

$$f_U(u) \propto u^{n-1} e^{-\psi(u)} \prod_{i=1}^{k} \tau_{n_i|x_i^*}(u) \, \mathbf{1}_{(0,\infty)}(u),$$

where ψ denotes the Laplace exponent of $\tilde{\mu}$,

$$\psi(u) = \int_{(0,\infty)\times\mathbb{X}} \left(1 - e^{-us}\right) \mathrm{d}\nu(s, x),$$

and

$$\tau_{m|x}(u) = \int_{(0,\infty)} e^{-us} s^m \, \mathrm{d}\rho_x(s).$$

We now state Theorem 11 from James et al. (2009).

Theorem 1.8. *Let $\tilde{\mu}$ be a CRM with Lévy intensity $\mathrm{d}\nu(s, x) = \mathrm{d}\rho_x(s)\, \mathrm{d}P_0(x)$, and let $x_1, \ldots, x_n$ be observations generated from model (1.1). Define $\tilde{\mu}^*$ such that*

$$\tilde{\mu}^* \mid U \stackrel{d}{=} \tilde{\mu}_U + \sum_{i=1}^{k} J_i^U \delta_{x_i^*},$$

where:

(i) $\tilde{\mu}_U$ is a CRM with Lévy intensity

$$\mathrm{d}\nu_U(s, x) = e^{-Us} \, \mathrm{d}\rho_x(s) \, \mathrm{d}P_0(x);$$

(ii) for each $i = 1, \dots, k$, the random variable J_i^U has density

$$f_i(s) \propto s^{n_i} e^{-Us} \, \mathrm{d}\rho_{x_i^*}(s);$$

(iii) the random variables $(J_i^U)_{i=1}^k$ are mutually independent and independent of $\tilde{\mu}_U$.

Then $\mathcal{L}(\tilde{\mu}^*)$ is a version of the posterior distribution of $\tilde{\mu}$ given $X_{1:n}$ under model (1.1).

Note that the posterior random measure is a completely random measure conditional on the random variable U. This observation motivates the notion of a Cox CRM, i.e., a random completely random measure.

Definition 1.9. Let $\tilde{\nu}$ be a random Lévy intensity with law F, and let $\mathcal{L}(\tilde{\mu}_\nu)$ denote the law of a CRM with fixed Lévy intensity ν.

A random measure $\tilde{\mu}$ is called a Cox CRM if, for all $A \in \mathcal{B}(\mathcal{M}_B(\mathbb{X}))$,

$$\mathbb{P}\big(\tilde{\mu} \in A\big) = \int_{\mathcal{M}((0,\infty)\times\mathbb{X})} \mathcal{L}(\tilde{\mu}_\nu)(A) \, \mathrm{d}F(\nu).$$

The statistical interpretation is that there exists a random Lévy intensity $\tilde{\nu}$ such that

$$\tilde{\mu} \mid \tilde{\nu} \sim \mathrm{CRM}(\tilde{\nu}).$$

The same identifiability issue arises for Cox CRMs as in the case of CRMs. The notion of scaling extends naturally to this setting and provides a representative element within each equivalence class of Cox CRMs that induce the same random probability measure through normalization.

Definition 1.10. Let $\tilde{\mu}$ be a Cox CRM with random Lévy intensity $\tilde{\nu} \sim F$, and suppose that F is concentrated on the set of Lévy intensities defining infinitely active CRMs with finite means. The random measure $\tilde{\mu}_S$ is a scaled Cox CRM if, for all $A \in \mathcal{B}(\mathcal{M}_B(\mathbb{X}))$,

$$\mathbb{P}\big(\tilde{\mu}_S \in A\big) = \int_{\mathcal{M}((0,\infty)\times\mathbb{X})} \mathcal{L}(\tilde{\mu}_{\nu,S})(A) \, \mathrm{d}F(\nu),$$

where $\tilde{\mu}_{\nu,S}$ denotes the scaled CRM associated with ν.

Under additional assumptions on the Lévy intensities (see Theorem 10 in Catalano and Lavenant (2023)), scaled Cox CRMs serve as representative elements of equivalence classes. In particular, for two Cox CRMs $\tilde{\mu}^1, \tilde{\mu}^2$, we have

$$\frac{\tilde{\mu}^1}{\tilde{\mu}^1(\mathbb{X})} \overset{d}{=} \frac{\tilde{\mu}^2}{\tilde{\mu}^2(\mathbb{X})} \quad \Longleftrightarrow \quad \tilde{\mu}^1_S \overset{d}{=} \tilde{\mu}^2_S.$$

We denote by $\tilde{\nu}_S$ the random Lévy intensity associated with the scaled Cox CRM. Let us now discuss the several examples of CRMs.

1.2.5 The Gamma Process and the Dirichlet Process

We begin this subsection by describing another derivation of Dirichlet process via completely random measures.

Example 1.11 (Gamma CRM)**.** A completely random measure is gamma if its Lévy intensity is

$$\mathrm{d}\nu(s,x) = \alpha \frac{e^{-bs}}{s} \mathrm{d}s \mathrm{d}P_0(x),$$

where $\alpha > 0$ is a total base measure, $b > 0$ is a scale parameter, and $P_0 \in \mathcal{P}(\mathbb{X})$ is a base probability.

The importance of the gamma CRM stems from the fact that its normalization yields a Dirichlet process. Indeed, define the random probability measure

$$\tilde{P} = \frac{\tilde{\mu}}{\tilde{\mu}(\mathbb{X})},$$

where $\tilde{\mu}$ is a gamma CRM. We now show that $\tilde{P}$ is a Dirichlet process with parameters α and P_0.

To this end, it is enough to verify that for any measurable partition $A_1, \ldots, A_n$ of $\mathbb{X}$,

$$\left(\tilde{P}(A_1), \ldots, \tilde{P}(A_n)\right) \sim \text{Dirichlet}\left(\alpha P_0(A_1), \ldots, \alpha P_0(A_n)\right).$$

Let $A \in \mathcal{B}(\mathbb{X})$. To obtain the law of $\tilde{\mu}(A)$, we compute its Laplace transform.

Define $f(x) = t\mathbf{1}_A(x)$ for $t > 0$. Then,

$$\begin{aligned}
\mathbb{E}\left[e^{-t\tilde{\mu}(A)}\right] &= \mathbb{E}\left[e^{-\int_{\mathbb{X}} f(x)\mathrm{d}\tilde{\mu}(x)}\right] \\
&= \exp\left\{-\int_{(0,\infty)\times\mathbb{X}} \left(1 - e^{-sf(x)}\right) \mathrm{d}\nu(s,x)\right\} \\
&= \exp\left\{-\alpha P_0(A) \int_0^\infty \left(1 - e^{-st}\right) \frac{e^{-bs}}{s} \mathrm{d}s\right\} \\
&= \exp\left\{-\alpha P_0(A) \int_0^\infty \frac{e^{-bs} - e^{-(b+t)s}}{s} \mathrm{d}s\right\} \\
&= \exp\left\{-\alpha P_0(A) \log\left(\frac{b+t}{b}\right)\right\} \\
&= \left(\frac{b}{b+t}\right)^{\alpha P_0(A)}.
\end{aligned}$$

The last equality follows from the Frullani integral. Hence, this coincides with the Laplace transform of the Gamma$(\alpha P_0(A), b)$ random variable. Therefore, for any measurable partition $A_1, \ldots, A_n$,

$$\tilde{\mu}(A_i) \sim \text{Gamma}\left(\alpha P_0(A_i), b\right).$$

Since these random variables are independent and $\tilde{\mu}(\mathbb{X}) = \sum_{i=1}^n \tilde{\mu}(A_i)$, we obtain the desired conclusion.

1.2.6 The Pitman–Yor Process and Equivalences

Another widely used example of a random probability measure is the Pitman–Yor process. It can be defined in several ways: through exchangeable partition probability functions, stick-breaking processes, Poisson–Kingman processes, and CRMs. Here, we provide two definitions using the latter two approaches and show the equivalence between them.

We start by defining the Pitman–Yor process as a special case of a Poisson–Kingman process. Denote by $\Delta^{\downarrow} = \{(s_1, s_2, \ldots) : 1 \geq s_1 \geq s_2 \geq \cdots \geq 0, \quad \sum_{i=1}^{\infty} s_i = 1\}$. the space of decreasing sequences in $[0, 1]$.

Definition 1.12. Let $\mathcal{N}$ be a Poisson random measure on $(0, \infty)$ with Lévy intensity ρ. Let (J_i) be its random atoms, consider $J_{(1)} \geq J_{(2)} \geq \ldots$, and let $T = \sum_{i=1}^{\infty} J_i$. Then, denote by $\text{PK}(\rho \mid t)$ the conditional distribution of the random vector

$$\left(\frac{J_{(1)}}{T}, \frac{J_{(2)}}{T}, \ldots\right) \in \Delta^{\downarrow}$$

given $T = t$, and by $\text{PK}(\rho)$ the distribution of the same vector without conditioning.

For a probability measure η on $(0, \infty)$, the Poisson–Kingman distribution $\text{PK}(\rho, \eta)$ is the law on $\Delta^{\downarrow}$ defined as

$$\int_0^{\infty} \text{PK}(\rho \mid t)\, \mathrm{d}\eta(t).$$

The corresponding random probability measure on $\mathbb{X}$,

$$\tilde{P} = \sum_{i \geq 1} \frac{J_i}{T} \delta_{X_i},$$

where $\left(\frac{J_i}{T}\right)_{i \geq 1} \sim \text{PK}(\rho, \eta)$ and $X_1, X_2, \ldots \overset{\text{i.i.d}}{\sim} P_0 \in \mathcal{P}(\mathbb{X})$, is called a Poisson–Kingman process.

Definition 1.13. A Pitman–Yor process is a Poisson–Kingman process $\text{PK}(\rho, \eta)$, where

$$\mathrm{d}\rho(s) = \frac{\sigma}{\Gamma(1-\sigma)} s^{-1-\sigma} \mathrm{d}s, \quad \mathrm{d}\eta(t) = \frac{\sigma \Gamma(\theta)}{\Gamma\left(\frac{\theta}{\sigma}\right)} t^{-\theta} f(t) \mathrm{d}t \quad \text{for } \sigma \in (0, 1), \theta > 0,$$

and f is a probability density with Laplace transform $\lambda \mapsto e^{-\lambda^\sigma}$.

The function ρ with the form given in Definition 1.13 is referred to as the intensity function of the σ-stable process. In particular, the σ-stable process is defined as follows.

Definition 1.14. $\tilde{\mu}$ is a σ-stable completely random measure if the jump part of the Lévy intensity has the form

$$\mathrm{d}\rho(s) = \frac{\sigma}{\Gamma(1-\sigma)} s^{-1-\sigma} \mathrm{d}s,$$

for some $\sigma \in (0,1)$.

Another way to define the Pitman–Yor process is via normalization of a Cox completely random measure.

Definition 1.15. Let $\tilde{\mu}$ be a Cox CRM$(\tilde{\nu})$ with random Lévy intensity

$$\mathrm{d}\tilde{\nu}_\xi(s,x) = \frac{\xi\sigma}{\Gamma(1-\sigma)} \frac{e^{-s}}{s^{1+\sigma}} \mathrm{d}s \, \mathrm{d}P_0(x),$$

where $\xi \sim \Gamma\left(\frac{\theta}{\sigma}, 1\right)$ with density $g(\xi) = \frac{1}{\Gamma(\frac{\theta}{\sigma})} \xi^{\frac{\theta}{\sigma}-1} e^{-\xi}$, $\theta \in (0,\infty)$ and $\sigma \in (0,1)$. The corresponding normalized random probability measure,

$$\tilde{P} = \frac{\tilde{\mu}}{\tilde{\mu}(\mathbb{X})},$$

is called the Pitman–Yor process.

We are now ready to show that Definition 1.15 and Definition 1.13 define the same random probability measures.

Since we work with homogeneous CRMs, our strategy is to show that the distribution of the normalized ranked jumps induced by the Cox completely random measure of Definition 1.15 is PK(ρ,η), where ρ and η are as in Definition 1.13. To this end, we first reduce the Cox CRM to a simpler form. The motivation is that, for fixed ξ, the associated Lévy intensity resembles the exponential tilting of a σ-stable CRM.

- Step 1: Reduction of the Cox CRM.

Denote by $\tilde{\nu}^1_\xi$ and $\tilde{\nu}^2_\xi$ the random Lévy intensities defined by

$$\mathrm{d}\tilde{\nu}^1_\xi(s,x) = \frac{\xi\sigma}{\Gamma(1-\sigma)} s^{-1-\sigma} e^{-s} \, \mathrm{d}s \, \mathrm{d}P_0(x), \qquad \mathrm{d}\tilde{\nu}^2_\xi(s,x) = \frac{\sigma}{\Gamma(1-\sigma)} s^{-1-\sigma} e^{-\xi^{\frac{1}{\sigma}} s} \, \mathrm{d}s \, \mathrm{d}P_0(x),$$

and let $\tilde{\mu}^1$ and $\tilde{\mu}^2$ denote the corresponding Cox CRMs. Similarly, define the Lévy measures

$$\mathrm{d}\rho^1_\xi(s) = \frac{\xi\sigma}{\Gamma(1-\sigma)} s^{-1-\sigma} e^{-s} \, \mathrm{d}s, \qquad \mathrm{d}\rho^2_\xi(s) = \frac{\sigma}{\Gamma(1-\sigma)} s^{-1-\sigma} e^{-\xi^{\frac{1}{\sigma}} s} \, \mathrm{d}s.$$

For simplicity of notation, we use ρ^i to denote both the density and the associated measure. Fix $\xi > 0$ and note that $\rho^2_\xi(s) = \xi^{\frac{1}{\sigma}}\rho^1_\xi(\xi^{\frac{1}{\sigma}}s)$. By Lemma SM3 in Catalano and Lavenant (2023),

$$\tilde{\mu}^2 \stackrel{d}{=} \xi^{-\frac{1}{\sigma}}\tilde{\mu}^1.$$

Thus, for fixed ξ, both CRMs induce the same random probability measure, and consequently the two Cox CRMs also induce the same random probability measure.

- Step 2: Matching the distributions of normalized ranked jumps.

Working with the reduced Cox CRM, define

$$\mathrm{d}\tilde{\nu}_\xi(s,x) = \frac{\sigma}{\Gamma(1-\sigma)}\, s^{-1-\sigma}e^{-\xi^{\frac{1}{\sigma}}s}\,\mathrm{d}s\,\mathrm{d}P_0(x), \qquad \mathrm{d}\rho_\xi(s) = \frac{\sigma}{\Gamma(1-\sigma)}\, s^{-1-\sigma}e^{-\xi^{\frac{1}{\sigma}}s}\,\mathrm{d}s,$$

and let $\tilde{\mu}$ denote the associated Cox CRM.

Let $\tilde{\gamma}$ denote the distribution of the normalized ranked jumps from $\tilde{\mu}$. By definition, for every $A \in \mathcal{B}(\Delta^{\downarrow})$,

$$\tilde{\gamma}(A) = \int_0^\infty \mathrm{PK}(\rho_\xi)(A)\, g(\xi)\,\mathrm{d}\xi,$$

where g is the density of $\mathrm{Gamma}\big(\frac{\theta}{\sigma},1\big)$.

Disintegrating $\mathrm{PK}(\rho_\xi)$ with respect to the distribution of the total jump sum gives

$$\tilde{\gamma}(A) = \int_0^\infty \int_0^\infty \mathrm{PK}(\rho_\xi \mid t)(A)\, f_\xi(t)\,\mathrm{d}t\, g(\xi)\,\mathrm{d}\xi,$$

where f_ξ is the density of the total jump sum of $\mathrm{CRM}(\tilde{\nu}_\xi)$ for fixed ξ, and $\mathrm{PK}(\rho_\xi \mid t)$ denotes the conditional distribution of the normalized ranked jumps given total jump sum t.

Since ρ_ξ is the exponential tilting of the Lévy measure of a σ-stable CRM, we have $\mathrm{PK}(\rho_\xi \mid t) = \mathrm{PK}(\rho \mid t)$ (see Example 14.46 of Ghosal and Van der Vaart (2017)). Therefore,

$$\tilde{\gamma}(A) = \int_0^\infty \int_0^\infty \mathrm{PK}(\rho \mid t)(A)\, f_\xi(t)\,\mathrm{d}t\, g(\xi)\,\mathrm{d}\xi = \int_0^\infty \mathrm{PK}(\rho \mid t)(A)\left[\int_0^\infty f_\xi(t)\, g(\xi)\,\mathrm{d}\xi\right]\mathrm{d}t.$$

It remains to show that

$$\int_0^\infty f_\xi(t)\, g(\xi)\,\mathrm{d}\xi = \frac{\sigma\,\Gamma(\theta)}{\Gamma\big(\frac{\theta}{\sigma}\big)}\, t^{-\theta}\, f(t),$$

where f is the density of the total jump sum of a σ-stable CRM. The density f is characterized by its Laplace transform $\lambda \mapsto e^{-\psi(\lambda)}$ with $\psi(\lambda) = \lambda^\sigma$ (see Example 14.43 in Ghosal and Van der Vaart (2017)). By Example 14.46 in Ghosal and Van der Vaart (2017),

$$f_\xi(t) = f(t)\, e^{\psi\left(\xi^{\frac{1}{\sigma}}\right) - \xi^{\frac{1}{\sigma}} t} = f(t)\, e^{\xi - \xi^{\frac{1}{\sigma}} t}.$$

Hence,

$$\begin{aligned}
\int_0^\infty f_\xi(t)\, g(\xi)\, \mathrm{d}\xi &= \int_0^\infty f(t)\, e^{\xi - \xi^{\frac{1}{\sigma}} t} \frac{1}{\Gamma\left(\frac{\theta}{\sigma}\right)} \xi^{\frac{\theta}{\sigma}-1} e^{-\xi}\, \mathrm{d}\xi \\
&= \frac{f(t)}{\Gamma\left(\frac{\theta}{\sigma}\right)} \int_0^\infty \xi^{\frac{\theta}{\sigma}-1} e^{-\xi^{\frac{1}{\sigma}} t}\, \mathrm{d}\xi \\
&= \frac{\sigma\, \Gamma(\theta)}{\Gamma\left(\frac{\theta}{\sigma}\right)} t^{-\theta} f(t),
\end{aligned}$$

where the last equality follows from the substitution $\tau = \xi^{\frac{1}{\sigma}}$.

1.3 Distances Between Laws of Random Measures and RPMs

In this section, we overview several ways to define distances on the laws of random probability measures and, more generally, on random measures. The distances on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ that are mainly used for two-sample testing are Wasserstein over Wasserstein (WoW) and the Hierarchical Integral Probability Metric (HIPM). The other distances are defined on $\mathcal{M}_B(\mathbb{X})$ and are used to study the merging rate of opinions.

1.3.1 Distances on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$

We consider two distance functions on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$. The first, referred to as Wasserstein over Wasserstein, is the natural lifting of the Wasserstein distance on $\mathcal{P}(\mathbb{X})$; it has been widely used in Bayesian nonparametrics Nguyen (2016) and machine learning Ho et al. (2017); Dukler et al. (2019). The second, termed the Hierarchical Integral Probability Metric, is a very recent development defined in Catalano and Lavenant (2024). Let us start with the definition of Wasserstein over Wasserstein.

Only in this subsection and in the chapter on two-sample testing, we assume that the sample space $\mathbb{X}$ is Polish with bounded diameter. Using the distance function $d_{\mathbb{X}}$ on $\mathbb{X}$, one can define the Wasserstein distance on $\mathcal{P}(\mathbb{X})$ by

$$\mathcal{W}_1(P^1, P^2) = \inf_{\pi \in \Gamma(P^1, P^2)} \int_{\mathbb{X} \times \mathbb{X}} d_{\mathbb{X}}(x, y)\, \pi(dx, dy), \qquad P^1, P^2 \in \mathcal{P}(\mathbb{X}),$$

where

$$\Gamma(P^1, P^2) = \left\{\pi \in \mathcal{P}(\mathbb{X}^2) : \pi(A \times \mathbb{X}) = P^1(A),\ \pi(\mathbb{X} \times A) = P^2(A) \quad \forall A \in \mathcal{B}(\mathbb{X})\right\}.$$

Note that this is the Wasserstein distance of order 1. Since we do not consider other Wasserstein distances, we denote it simply by $\mathcal{W}$ instead of $\mathcal{W}_1$.

In general, $\mathcal{W}$ is a distance function on the Wasserstein space of order 1, defined as

$$\mathcal{P}_1(\mathbb{X}) = \left\{P \in \mathcal{P}(\mathbb{X}) : \int_{\mathbb{X}} d_{\mathbb{X}}(x, x_0) P(dx) < \infty, \quad x_0 \in \mathbb{X}\right\}.$$

Since we assume that $\mathbb{X}$ has bounded diameter, all $P \in \mathcal{P}(\mathbb{X})$ belong to $\mathcal{P}_1(\mathbb{X})$. The assumption that $\mathbb{X}$ is Polish and $d_{\mathbb{X}}$ is bounded implies that the metric space $(\mathcal{P}(\mathbb{X}), \mathcal{W})$ is also Polish; see Remark 7.1.7 in Luigi et al. (2008). Note also that $(\mathcal{P}(\mathbb{X}), \mathcal{W})$ has bounded diameter, since $d_{\mathbb{X}}$ is bounded. Therefore, instead of defining a distance function on $\mathcal{P}_1(\mathcal{P}(\mathbb{X}))$, we can define a distance function on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ using $\mathcal{W}$ as the ground metric.

Definition 1.16 (Wasserstein over Wasserstein)**.** Given $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, the Wasserstein over Wasserstein (WoW) distance between Q^1 and Q^2 is

$$\mathcal{W}_{\mathcal{W}}(Q^1, Q^2) = \inf_{\pi \in \Gamma(Q^1, Q^2)} \int_{\mathcal{P}(\mathbb{X})^2} \mathcal{W}(P^1, P^2)\, \pi(dP^1, dP^2).$$

Here, $\Gamma(Q^1, Q^2)$ denotes the set of probability measures on $\mathcal{P}(\mathbb{X}) \times \mathcal{P}(\mathbb{X})$ with marginals Q^1 and Q^2. Again, Remark 7.1.7 in Luigi et al. (2008), together with the fact that $\mathcal{W}$ has bounded diameter, implies that the metric space $(\mathcal{P}(\mathcal{P}(\mathbb{X})), \mathcal{W}_{\mathcal{W}})$ is Polish with bounded diameter.

The second distance function we consider is HIPM, introduced in Catalano and Lavenant (2024). The primary intuition behind its definition is as follows. Let Q^1, Q^2 be probability measures on $\mathcal{P}(\mathbb{X})$, and let $\widetilde{P}^1 \sim Q^1$ and $\widetilde{P}^2 \sim Q^2$. We compute the distance between Q^1 and Q^2 by considering the distances between the random integrals $\widetilde{P}^1(f)$ and $\widetilde{P}^2(f)$, where f ranges over a class of functions from $\mathbb{X}$ to $\mathbb{R}$. By considering these projections, we shift the focus from $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ to $\mathcal{P}(\mathbb{R})$, the latter being significantly easier to handle. As noted in Catalano and Lavenant (2024), such a projection is justified by the fact that, for a random probability measure $\widetilde{P}$, the laws of the random integrals $\widetilde{P}(f)$, where f belongs to the set of continuous and bounded functions, are sufficient to characterize the law of $\widetilde{P}$. This leads to the following definition.

Definition 1.17 (Hierarchical integral probability metric)**.** Given $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, the hierarchical integral probability metric between Q^1 and Q^2 is defined as

$$d_{\mathcal{F}}(Q^1, Q^2) = \sup_{f \in \mathcal{F}} \mathcal{W}(\mathcal{L}(\widetilde{P}^1(f)), \mathcal{L}(\widetilde{P}^2(f))),$$

where $\widetilde{P}^k \sim Q^k$ for $k = 1, 2$ and $\mathcal{F}$ is a class of bounded and measurable functions from $\mathbb{X}$ to $\mathbb{R}$.

Alternatively, the HIPM can be expressed as

$$d_{\mathcal{F}}(Q^1, Q^2) = \sup_{f \in \mathcal{F}} \mathcal{W}(Q^1 \circ \hat{f}^{-1}, Q^2 \circ \hat{f}^{-1}),$$

where $\hat{f}$ denotes the lifting of the function f, i.e., $\hat{f}(P) = \int_{\mathbb{X}} f \, dP$, and $Q^k \circ \hat{f}^{-1}$ is the corresponding push-forward measure.

As long as the function class $\mathcal{F}$ is sufficiently rich, this construction defines a valid distance on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$. Hereafter, we set $\mathcal{F} = \text{Lip}_1(\mathbb{X}, \mathbb{R})$ and denote the corresponding metric $d_{\mathcal{F}}$ by d_{Lip}.

Both distance functions possess desirable topological properties. In particular, by Theorem 3.1 in Catalano and Lavenant (2024), WoW metrizes weak convergence on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$, and HIPM does the same when $\mathbb{X}$ is compact.

There is a relation between the two distance functions; specifically, they are special cases of an Integral Probability Metric defined on a generic Polish space.

Definition 1.18 (Integral Probability Metric)**.** Let $(\mathbb{Y}, d_{\mathbb{Y}})$ be a Polish space and $\mathcal{H}$ a class of bounded and measurable functions from $\mathbb{Y}$ to $\mathbb{R}$. The Integral Probability Metric (IPM) between $P^1, P^2 \in \mathcal{P}(\mathbb{Y})$ is

$$\mathcal{I}_{\mathcal{H}}(P^1, P^2) = \sup_{f \in \mathcal{H}} \left| \int_{\mathbb{Y}} f(y) P^1(dy) - \int_{\mathbb{Y}} f(y) P^2(dy) \right|.$$

First, let us show that WoW can be written as an integral probability metric. By Remark 6.5 in Villani et al. (2008), if P^1, P^2 belong to $\mathcal{P}_1(\mathbb{X})$, the Wasserstein space of order 1, then the Wasserstein distance can be written as

$$\mathcal{W}(P^1, P^2) = \sup_{f \in \text{Lip}_1(\mathbb{X}, \mathbb{R})} \left| \int_{\mathbb{X}} f(x) P^1(dx) - \int_{\mathbb{X}} f(x) P^2(dx) \right|,$$

where $\text{Lip}_1(\mathbb{X}, \mathbb{R})$ denotes the set of 1-Lipschitz continuous functions from $\mathbb{X}$ to $\mathbb{R}$. Since $(\mathcal{P}(\mathcal{P}(\mathbb{X})), \mathcal{W}_{\mathcal{W}})$ is a Polish space with bounded diameter, we can use the above result to obtain, for all $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$,

$$\mathcal{W}_{\mathcal{W}}(Q^1, Q^2) = \sup_{f \in \mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})} \left| \int_{\mathcal{P}(\mathbb{X})} f(P) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} f(P) Q^2(dP) \right|,$$

where $\mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})$ denotes the set of 1-Lipschitz continuous functions from $\mathcal{P}(\mathbb{X})$ to $\mathbb{R}$ with respect to $\mathcal{W}$.

Similarly, we show that HIPM can be written as an integral probability metric. For $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$,

$$\begin{aligned}
d_{\mathrm{Lip}}(Q^1, Q^2) &= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \mathcal{W}(Q^1 \circ \hat{f}^{-1}, Q^2 \circ \hat{f}^{-1}) \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \sup_{g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})} \left| \int_{\mathbb{R}} g(x)(Q^1 \circ \hat{f}^{-1})(dx) - \int_{\mathbb{R}} g(x)(Q^2 \circ \hat{f}^{-1})(dx) \right| \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \sup_{g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})} \left| \int_{\mathcal{P}(\mathbb{X})} g(\widehat{f}(P)) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} g(\widehat{f}(P)) Q^2(dP) \right| \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \sup_{g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})} \left| \int_{\mathcal{P}(\mathbb{X})} g(P(f)) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} g(P(f)) Q^2(dP) \right| \\
&= \sup_{h \in \mathcal{D}} \left| \int_{\mathcal{P}(\mathbb{X})} h(P) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} h(P) Q^2(dP) \right|,
\end{aligned}$$

where $\mathcal{D} = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)),\ g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R}), f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})\}$.

By representing both distance functions as Integral Probability Metrics, we see that HIPM is upper bounded by the WoW distance. Indeed, we can show that $\mathcal{D}$ is a subset of $\mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})$: let $h \in \mathcal{D}$ and $P^1, P^2 \in \mathcal{P}(\mathbb{X})$; then

$$\begin{aligned}
|h(P^1) - h(P^2)| = |g(P^1(f)) - g(P^2(f))| \leq |P^1(f) - P^2(f)| &\leq \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} |P^1(f) - P^2(f)| \\
&= \mathcal{W}(P^1, P^2).
\end{aligned}$$

1.3.2 Distances on $\mathcal{M}_B(\mathbb{X})$

Differently from the previous subsection, we only assume that $\mathbb{X}$ is a Polish space equipped with its Borel σ-algebra $\mathcal{B}(\mathbb{X})$. Our motivation for defining a distance between laws of random measures stems from the study of completely random measures. Since we will be working with CRMs taking values in the space of finite measures on $\mathbb{X}$, we begin by introducing a distance on the space $\mathcal{M}_B(\mathbb{X})$.

Recall that the space $\mathcal{M}_B(\mathbb{X})$ is endowed with the Borel σ-algebra generated by the topology of weak convergence. Analogously to the case of $\mathcal{P}(\mathbb{X})$, this σ-algebra coincides with the smallest σ-algebra that makes all maps of the form

$$\mu \mapsto \mu(f), \quad f \in C_b(\mathbb{X}),$$

measurable.

Since two measures in $\mathcal{M}_B(\mathbb{X})$ may have different total masses, the standard Wasserstein distance is not directly applicable. Instead, we consider the bounded Lipschitz distance.

Definition 1.19. Given $\mu^1, \mu^2 \in \mathcal{M}_B(\mathbb{X})$, the bounded Lipschitz distance BL is defined as

$$\mathrm{BL}(\mu^1, \mu^2) = \sup\left\{\int_{\mathbb{X}} f \mathrm{d}\mu^1 - \int_{\mathbb{X}} f \mathrm{d}\mu^2 \ : \ f \text{ is 1-Lipschitz and } \sup_{x\in\mathbb{X}} |f(x)| \leq 1\right\}.$$

By lifting the distance BL to the space of laws of random measures, denoted by $\mathcal{P}(\mathcal{M}_B(\mathbb{X}))$, we obtain a distance on $\mathcal{P}(\mathcal{M}_B(\mathbb{X}))$.

Definition 1.20. Given $Q^1, Q^2 \in \mathcal{P}(\mathcal{M}_B(\mathbb{X}))$, the distance $\mathcal{W}_{\mathrm{BL}}$ between them is defined as

$$\mathcal{W}_{\mathrm{BL}}(Q^1, Q^2) = \min_{\pi \in \Gamma(Q^1, Q^2)} \int_{\mathcal{M}_B(\mathbb{X})\times\mathcal{M}_B(\mathbb{X})} \mathrm{BL}(\mu^1, \mu^2) \, \mathrm{d}\pi(\mu^1, \mu^2),$$

where $\Gamma(Q^1, Q^2)$ denotes the set of probability measures on $\mathcal{M}_B(\mathbb{X}) \times \mathcal{M}_B(\mathbb{X})$ with marginals Q^1 and Q^2.

Remark 1.21. As mentioned in the previous subsection, the Wasserstein distance is, in general, a metric on the space of probability measures with finite first moments. In the present setting, since BL is bounded by 2, $\mathcal{W}_{\mathrm{BL}}$ is a well-defined distance on $\mathcal{P}(\mathcal{M}_B(\mathbb{X}))$.

If $\tilde{\mu}^1$ and $\tilde{\mu}^2$ are random measures with laws Q^1 and Q^2, respectively, we refer to $\mathcal{W}_{\mathrm{BL}}(\tilde{\mu}^1, \tilde{\mu}^2)$ as the distance between the random measures, with the understanding that this denotes the distance between their laws, i.e., $\mathcal{W}_{\mathrm{BL}}(Q^1, Q^2)$. For a more detailed overview of the properties of $\mathcal{W}_{\mathrm{BL}}$, see Catalano and Lavenant (2023).

1.3.3 Distance for Completely Random Measures

In the previous subsection, we defined a distance between laws of random measures taking values in the space of finite measures. Although this framework also includes completely random measures, the resulting distance is generally not tractable in this setting. Since our goal is to work with distances that can be explicitly computed or bounded, we now introduce a distance tailored specifically to CRMs.

As discussed earlier, a completely random measure is uniquely characterized by its Lévy intensity, which is a σ-finite measure on $(0,\infty)\times\mathbb{X}$. For this reason, we define the distance at the level of Lévy intensities. Note that, since Lévy intensities have infinite total mass, we cannot use the bounded Lipschitz distance. Recall that, given a Lévy intensity ν, its disintegration can be written as

$$\mathrm{d}\nu(s,x)=\mathrm{d}\rho_x(s)\mathrm{d}P_0(x),$$

where P_0 is a probability measure on $\mathbb{X}$ and, for each $x\in\mathbb{X}$, ρ_x is a measure on $(0,\infty)$. This decomposition suggests defining a discrepancy between two Lévy intensities by coupling atoms in $\mathbb{X}$ and comparing the corresponding measures on $(0,\infty)$ along such a coupling. To do so, we first need a distance on the space of measures on $(0,\infty)$.

Since the measures ρ_x may have infinite total mass, neither the standard Wasserstein distance nor the bounded Lipschitz distance is directly applicable. We therefore use the notion introduced in Figalli and Gigli (2010), which extends the Wasserstein distance to positive measures that may have infinite total mass.

Definition 1.22. Let ρ^1,ρ^2 be two positive Borel measures on $(0,\infty)$ with finite first moments about 0. The extended Wasserstein distance between ρ^1 and ρ^2 is defined by

$$\mathcal{W}_*(\rho^1,\rho^2)=\int_0^\infty |U_1(s)-U_2(s)|\mathrm{d}s,$$

where

$$U_i(s)=\rho^i((s,\infty)),\qquad s>0,$$

is the tail integral of ρ^i.

Having introduced distances on measures on $(0,\infty)$ and on the space $\mathbb{X}$, we can now define a discrepancy between Lévy intensities.

Definition 1.23. Let ν^1,ν^2 be two Lévy intensities such that $\mathcal{M}_1(\nu^i)<\infty$, and let

$$\mathrm{d}\nu^i(s,x)=\mathrm{d}\rho^i_x(s)\mathrm{d}P^i_0(x),\qquad i=1,2,$$

be their canonical decompositions. The adapted extended Wasserstein discrepancy between ν^1 and ν^2 is defined as

$$\mathrm{AD}(\nu^1,\nu^2)=\inf_{\pi\in\Gamma(P_0^1,P_0^2)}\int_{\mathbb{X}\times\mathbb{X}}\left[\frac{\mathcal{M}_1(\nu^1)+\mathcal{M}_1(\nu^2)}{2}d_{\mathbb{X},t}(x,y)+\mathcal{W}_*(\rho^1_x,\rho^2_y)\right]\mathrm{d}\pi(x,y),$$

where

$$d_{\mathbb{X},t}(x,y)=\min\{d_{\mathbb{X}}(x,y),2\}$$

denotes the truncated metric on $\mathbb{X}$.

The use of the truncated metric $d_{\mathbb{X},t}$ instead of the original metric $d_{\mathbb{X}}$ is motivated by the bounds developed later. More precisely, our goal is to upper bound Wasserstein over bounded Lipschitz by the discrepancy AD. One of the key ingredients in that argument is the fact that the bounded Lipschitz distance is upper bounded by the Wasserstein distance on $\mathcal{P}(\mathbb{X})$. Working with the truncated metric leads to sharper bounds.

It is also worth noting that AD is not a distance. Indeed, the factor

$$\frac{\mathcal{M}_1(\nu^1) + \mathcal{M}_1(\nu^2)}{2}$$

makes the triangle inequality fail. Thus, AD should be regarded as a discrepancy rather than a distance. The following remark is stated in Catalano and Lavenant (2023) (see Remark 2 on page 10). Due to its importance for the computations involved in merging, we restate it here.

Remark 1.24. Consider two completely random measures with Lévy intensities ν^1, ν^2 of the form

$$\mathrm{d}\nu^1(s,x) = \mathrm{d}\rho^1_x(s)\mathrm{d}P^1_0(x), \qquad \mathrm{d}\nu^2(s,x) = \mathrm{d}\rho^2(s)\mathrm{d}P^2_0(x),$$

that is, one of them is a homogeneous CRM. Then the adapted extended Wasserstein discrepancy between ν^1 and ν^2 decomposes as the sum of a scaled distance between the laws of the atoms and the expected distance between the laws of the jumps, i.e.,

$$\mathrm{AD}(\nu^1, \nu^2) = \frac{M_1(\nu^1) + M_1(\nu^2)}{2} \mathcal{W}(P^1_0, P^2_0) + \mathbb{E}_{X \sim P^1_0}\left[\mathcal{W}_*(\rho^1_X, \rho^2)\right].$$

We now discuss the distance between Cox CRMs. The motivation stems from the fact that, posterior distributions are random completely random measures, namely Cox CRMs. Since Cox CRMs are characterized by random Lévy intensities, it is natural to define a distance at the level of their laws by lifting the adapted extended Wasserstein discrepancy.

More precisely, let $\tilde{\nu}^1, \tilde{\nu}^2$ be random Lévy intensities. We define

$$\mathcal{W}_{\mathrm{AD}}(\mathcal{L}(\tilde{\nu}^1), \mathcal{L}(\tilde{\nu}^2)) = \inf_{\pi \in \Gamma(\mathcal{L}(\tilde{\nu}^1), \mathcal{L}(\tilde{\nu}^2))} \mathbb{E}_{(\nu^1,\nu^2)\sim\pi}\left[\mathrm{AD}(\nu^1, \nu^2)\right].$$

Finally, since in the case of merging rates our goal is to consider CRMs (or Cox CRMs) that uniquely determine a random probability measure through normalization, we restrict attention to scaled CRMs and Cox CRMs. In this case, the first moment of the Lévy intensity $\mathcal{M}_1(\nu)$ is equal to 1. As a consequence, by Proposition 5 in Catalano and Lavenant (2023), the quantity $\mathcal{W}_{\mathrm{AD}}$ defines a distance on the corresponding laws.

Our main tool for controlling the Wasserstein over bounded Lipschitz distance is provided by Theorem 6 in Catalano and Lavenant (2023), which states that, in the case of scaled Cox CRMs $\tilde{\mu}^1, \tilde{\mu}^2$ with random Lévy intensities $\tilde{\nu}^1, \tilde{\nu}^2$,

$$\mathcal{W}_{\mathrm{BL}}\big(\mathcal{L}\left(\tilde{\mu}^1\right), \mathcal{L}\left(\tilde{\mu}^1\right)\big) \leq \mathcal{W}_{\mathrm{AD}}(\mathcal{L}(\tilde{\nu}^1), \mathcal{L}(\tilde{\nu}^2)).$$

Chapter 2

Two-sample test for laws of random probabilities

2.1 Introduction

The two-sample test is a fundamental statistical procedure used to test the null hypothesis that two datasets are generated from the same underlying distribution. Specifically, given two sets of samples, the goal is to assess whether their associated distributions are statistically indistinguishable. The hypotheses are formulated as follows:

- Null hypothesis (H_0): The two distributions are identical.
- Alternative hypothesis (H_1): The two distributions are distinct.

Classic examples include the Kolmogorov–Smirnov test, which compares empirical cumulative distribution functions (CDFs) derived from independent and identically distributed (i.i.d.) samples. Other well-known univariate methods include the classical t-test and the Wilcoxon rank-sum test. As modern applications frequently involve multivariate observations in $\mathbb{R}^d$, several extensions have been developed, including kernel-based methods Gretton et al. (2012) and approaches based on the Wasserstein distance Ramdas et al. (2017); Wang et al. (2021).

Increasingly, however, data objects have become more complex, often appearing as probability measures themselves. Settings where observed samples are modeled as probability measures are now prevalent Petersen et al. (2022); examples include mortality rates across different countries, income distributions within various regions, and distributional summaries of functional connectivity in the brain. It is important to mention that, in the majority of cases, these probability measures are latent and are instead estimated from the raw observations they generate. Two-sample testing, in the case of probability measures as samples, naturally motivates the study of the laws of random probability measures, that is, probability measures defined on the space of probability measures.

This problem was previously addressed in Dubey and Müller (2019) using Fréchet means and variances. However, that approach has limitations: specifically, if the laws are not uniquely characterized by their Fréchet means and variances, the test may fail to provide a high rejection rate when the null hypothesis is false. In contrast, we adopt a fully nonparametric perspective, making no distributional assumptions on the underlying laws of the RPMs. Furthermore, theoretical guarantees are provided directly in terms of the raw observations used to estimate the underlying probability measures.

In this work, we adopt a distance-based approach. This framework first requires a metric to quantify the distance between two datasets. The core intuition is that if the observed distance is sufficiently "large," it is "unlikely" that the underlying distributions are the same. To formalize the terms "large" and "unlikely," one must determine a threshold based on a pre-specified significance level θ. This threshold is defined such that the probability of the observed distance exceeding it under the null hypothesis is at most θ. Consequently, the testing procedure is to reject the null hypothesis if the observed distance exceeds this threshold.

Our objective is to establish theoretical guarantees for both the false positive rate (the probability of incorrectly rejecting the null hypothesis when it is true) and the power (the probability of correctly rejecting the null hypothesis when it is false).

The rest of the paper is organized as follows. In Section 2.2, we formalize the sampling scheme and discuss its connections to exchangeability. Section 2.3 reviews existing distance functions on laws of random probability measures. In Section 2.4, we develop a theoretical threshold based on Rademacher complexity and illustrate its practical limitations through simulations. As a remedy, in Section 2.5, we introduce a practical threshold based on a permutation approach and establish its theoretical guarantees. Finally, in Section 2.6, we benchmark our method against existing approaches in the literature on simulated datasets and apply it to a mortality dataset.

2.2 Hierarchical sampling and exchangeability

We assume that the space $\mathbb{X}$ on which probability measures are defined is a Polish space (i.e., a complete, separable metric space) with a bounded diameter, meaning $\mathrm{diam}(\mathbb{X}) = \sup_{x,y\in\mathbb{X}} d_{\mathbb{X}}(x, y) < \infty$. Let $\mathcal{P}(\mathbb{X})$ denote the space of probability measures on $\mathbb{X}$. We will work with Random Probability Measures (RPMs), which are random variables taking values in $\mathcal{P}(\mathbb{X})$. The laws of RPMs are objects in $\mathcal{P}(\mathcal{P}(\mathbb{X}))$, which denotes the space of probability measures on $\mathcal{P}(\mathbb{X})$. Unless stated otherwise, the σ-algebra on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ is the Borel σ-algebra generated by the topology of weak convergence, where the continuous functions on $\mathcal{P}(\mathbb{X})$ are defined through the weak convergence on $\mathcal{P}(\mathbb{X})$.

In practice, it is rare to observe the probability measures themselves directly, especially when dealing with continuous probability distributions. Instead, one obtains observations from each latent probability measure. Therefore, we assume that given a law of a random probability measure $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, instead of observing $\widetilde{P}_1, \ldots, \widetilde{P}_n$ from Q, we observe samples from each $\widetilde{P}_i$. Such a collection of samples is referred to as a hierarchical sample:

Definition 2.1 (Hierarchical sample)**.** Given $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, the hierarchical sample refers to the collection of random variables $\{X_{i,j} : i = 1, \ldots, n,\ j = 1, \ldots, m\}$, sampled as

$$\begin{pmatrix} X_{1,1} & X_{1,2} & \cdots & X_{1,m} \mid \widetilde{P}_1 \overset{\text{i.i.d.}}{\sim} \widetilde{P}_1 \\ X_{2,1} & X_{2,2} & \cdots & X_{2,m} \mid \widetilde{P}_2 \overset{\text{i.i.d.}}{\sim} \widetilde{P}_2 \\ \vdots & \vdots & \ddots & \vdots \\ X_{n,1} & X_{n,2} & \cdots & X_{n,m} \mid \widetilde{P}_n \overset{\text{i.i.d.}}{\sim} \widetilde{P}_n \end{pmatrix},$$

where $\widetilde{P}_1, \ldots, \widetilde{P}_n \overset{\text{i.i.d.}}{\sim} Q$.

Throughout this paper, n and m will refer to the number of rows and columns of the hierarchical sample, respectively. We denote the hierarchical sample by $\mathbf{X}_{n,m} = (X_{i,j})$, where boldface letters refer to matrices and non-bold letters refer to their individual elements. With a slight abuse of notation, we will also sometimes denote $\mathbf{X}_{n,m} \sim Q$. Hierarchical samples are what we observe to decide whether two laws of random probability measures are the same. Before proceeding, let us discuss the limitations of the hierarchical sample.

Let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Suppose that we observe two hierarchical samples $\mathbf{X}_{n,m} \sim Q^1$ and $\mathbf{Y}_{n,m} \sim Q^2$. Note that each of the n sequences of length m from the first hierarchical sample consists of i.i.d. random variables from the following law on $\mathbb{X}^m$: for $i = 1, \ldots, n$,

$$P^{1,(m)}(A_1 \times \cdots \times A_m) := \mathbb{P}(X_{i,1} \in A_1, \ldots, X_{i,m} \in A_m) = \int_{\mathcal{P}(\mathbb{X})} \prod_{j=1}^{m} P(A_j) Q^1(dP).$$

Similarly, for the second hierarchical sample, we have

$$P^{2,(m)}(A_1 \times \cdots \times A_m) := \mathbb{P}(Y_{i,1} \in A_1, \ldots, Y_{i,m} \in A_m) = \int_{\mathcal{P}(\mathbb{X})} \prod_{j=1}^{m} P(A_j) Q^2(dP).$$

Such a characterization implies that the sequences $X_{i,1}, \ldots, X_{i,m}$ and $Y_{i,1}, \ldots, Y_{i,m}$ are finitely exchangeable. This means that each of the hierarchical samples, $\mathbf{X}_{n,m}$ and $\mathbf{Y}_{n,m}$, contains n finitely exchangeable sequences that are independent of one another. It is a well-known fact in Bayesian nonparametrics that a finite exchangeable sequence is not always in one-to-one correspondence with the law of a random probability measure. An implication of this is that it might happen that the laws of the RPMs are different, yet the hierarchical samples we observe are statistically indistinguishable; that is, all the finite exchangeable sequences we observe have the same probability distribution.

Indeed, consider two distinct Dirichlet processes

$$Q^1 = DP(\alpha_1, P_0), \quad Q^2 = DP(\alpha_2, P_0),$$

with $\alpha_1 \neq \alpha_2$. Then $Q^1 \neq Q^2$. Suppose that the size of the hierarchical samples from Q^1 and Q^2 is $n \times 1$. Then, the laws characterizing the two hierarchical samples are the same; i.e., for all $A \in \mathcal{B}(\mathbb{X})$,

$$\begin{aligned} P^{1,(1)}(A) &= \mathbb{P}(X_{i,1} \in A) = \int_{\mathcal{P}(\mathbb{X})} P(A)\, Q^1(dP), \\ P^{2,(1)}(A) &= \mathbb{P}(Y_{i,1} \in A) = \int_{\mathcal{P}(\mathbb{X})} P(A)\, Q^2(dP) = P_0(A). \end{aligned}$$

In this case, no testing procedure can achieve a low Type II error, which is the probability of failing to reject the null hypothesis when it is false. To overcome this limitation, we must ensure that

$$Q^1 = Q^2 \quad \text{if and only if} \quad P^{1,(m)} = P^{2,(m)}.$$

This indeed holds true for any Q^1, Q^2 for a sufficiently large m (see Lemma **??**).

As noted above, hierarchical samples $\mathbf{X}_{n,m} \sim Q^1$ and $\mathbf{Y}_{n,m} \sim Q^2$ are used to decide whether $Q^1 = Q^2$. We aim to develop a testing scheme inspired by the Kolmogorov-Smirnov (KS) test. Recall that in the KS test, one rejects the null hypothesis if the distance between the empirical measures of the samples exceeds a given threshold. In Section 2.3, we will equip the space $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ with a distance function; for the moment, assume that d defines a distance on that space. Then, we could naturally extend the idea of the KS test to our setting if we had access to the samples $\widetilde{P}^1_1, \dots, \widetilde{P}^1_n \overset{\text{i.i.d.}}{\sim} Q^1$ and $\widetilde{P}^2_1, \dots, \widetilde{P}^2_n \overset{\text{i.i.d.}}{\sim} Q^2$, by rejecting H_0 if

$$d\left(\frac{1}{n}\sum_{i=1}^{n} \delta_{\widetilde{P}^1_i}, \frac{1}{n}\sum_{i=1}^{n} \delta_{\widetilde{P}^2_i}\right)$$

exceeds some threshold. However, recall that we do not observe $\widetilde{P}^1_i$ and $\widetilde{P}^2_i$. Nevertheless, we can estimate them from the observations they generated; that is, we estimate each of them using the corresponding sequences $X_{i,1}, \dots, X_{i,m}$ and $Y_{i,1}, \dots, Y_{i,m}$ as

$$\widehat{P}^1_{i,m} = \frac{1}{m}\sum_{j=1}^{m} \delta_{X_{i,j}}, \quad \widehat{P}^2_{i,m} = \frac{1}{m}\sum_{j=1}^{m} \delta_{Y_{i,j}}, \quad i = 1, \dots, n.$$

Thus, the idea of the test is to reject H_0 if

$$d\left(\frac{1}{n}\sum_{i=1}^{n}\delta_{\widehat{P}^1_{i,m}}, \frac{1}{n}\sum_{i=1}^{n}\delta_{\widehat{P}^2_{i,m}}\right)$$

is higher than some threshold.

Note that the law of each $\widehat{P}^k_{i,m}$ is different from Q^k for $k = 1, 2$. Since these random probability measures are the main objects of study in this paper, we introduce a specific notation for their laws.

In particular, given $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, we denote by $Q^{(m)}$ the law of the empirical measure constructed from an exchangeable sequence; i.e.,

$$Q^{(m)} = \mathcal{L}\left(\frac{1}{m}\sum_{j=1}^{m}\delta_{Z_j}\right),$$

where $Z_1, \dots, Z_m$ is an exchangeable sequence generated as

$$\begin{aligned} P &\sim Q, \\ Z_1, \dots, Z_m \mid P &\overset{\text{i.i.d.}}{\sim} P. \end{aligned}$$

To summarize, given Q^1 and Q^2, since we estimate the random probability measures $\widehat{P}^k_{i,m}$ from the hierarchical samples $\mathbf{X}_{n,m}$ and $\mathbf{Y}_{n,m}$, we effectively observe

$$\widehat{P}^1_{1,m}, \dots, \widehat{P}^1_{n,m} \overset{\text{i.i.d.}}{\sim} Q^{1,(m)}, \quad \widehat{P}^2_{1,m}, \dots, \widehat{P}^2_{n,m} \overset{\text{i.i.d.}}{\sim} Q^{2,(m)}.$$

We aim to determine whether Q^1 and Q^2 are identical by comparing the empirical measures concentrated on these two samples. These empirical measures are formally defined as follows:

Definition 2.2 (Hierarchical estimator)**.** Given $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, and a hierarchical sample $\mathbf{X}_{n,m}$, the hierarchical estimator of Q is defined as

$$Q_{n,m} = \frac{1}{n}\sum_{i=1}^{n}\delta_{\widehat{P}_{i,m}},$$

where $\widehat{P}_{i,m} = \frac{1}{m}\sum_{j=1}^{m}\delta_{X_{i,j}}$ for all $i = 1, \dots, n$.

The testing scheme we develop will reject the null hypothesis whenever the hierarchical estimators $Q^1_{n,m}$ and $Q^2_{n,m}$ are far apart. To clarify what is meant by "far apart," we require a distance function on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$.

2.3 Distances between laws of random probabilities

In this paper, we consider two distance functions on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$. The first, referred to as "Wasserstein over Wasserstein," is the natural lifting of the Wasserstein distance on $\mathcal{P}(\mathbb{X})$; it has been widely used in Bayesian nonparametrics Nguyen (2016) and machine learning Ho et al. (2017); Dukler et al. (2019). The second, termed the "Hierarchical Integral Probability Metric," is a very recent development defined in Catalano and Lavenant (2024). Let us start with the definition of Wasserstein over Wasserstein.

Recall that the sample space $\mathbb{X}$ is Polish with a bounded diameter. Using the distance function $d_{\mathbb{X}}$ on $\mathbb{X}$, one can define the Wasserstein distance on $\mathcal{P}(\mathbb{X})$ by

$$\mathcal{W}_1(P^1, P^2) = \inf_{\pi \in \Gamma(P^1, P^2)} \int_{\mathbb{X}\times\mathbb{X}} d_{\mathbb{X}}(x, y)\, \pi(dx, dy), \qquad P^1, P^2 \in \mathcal{P}(\mathbb{X}),$$

where

$$\Gamma(P^1, P^2) = \left\{\pi \in \mathcal{P}(\mathbb{X}^2) : \pi(A \times \mathbb{X}) = P^1(A),\ \pi(\mathbb{X} \times A) = P^2(A) \quad \forall A \in \mathcal{B}(\mathbb{X})\right\}.$$

Note that this is the Wasserstein distance of order 1. Since we do not consider other Wasserstein distances, we denote it simply by $\mathcal{W}$ instead of $\mathcal{W}_1$.

In general, $\mathcal{W}$ is a distance function on the Wasserstein space of order 1, defined as

$$\mathcal{P}_1(\mathbb{X}) = \left\{P \in \mathcal{P}(\mathbb{X}) : \int_{\mathbb{X}} d_{\mathbb{X}}(x, x_0) P(dx) < \infty, \quad x_0 \in \mathbb{X}\right\}.$$

Since we assume that $\mathbb{X}$ has a bounded diameter, all $P \in \mathcal{P}(\mathbb{X})$ belong to $\mathcal{P}_1(\mathbb{X})$. The assumption that $\mathbb{X}$ is Polish and $d_{\mathbb{X}}$ is bounded implies that the metric space $(\mathcal{P}(\mathbb{X}), \mathcal{W})$ is also Polish (see Remark 7.1.7 in Luigi et al. (2008)). Note also that $(\mathcal{P}(\mathbb{X}), \mathcal{W})$ has a bounded diameter, as $d_{\mathbb{X}}$ is bounded. Therefore, instead of defining a distance function on $\mathcal{P}_1(\mathcal{P}(\mathbb{X}))$, we can define a distance function on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ using $\mathcal{W}$ as the ground metric:

Definition 2.3 (Wasserstein over Wasserstein)**.** Given $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, the Wasserstein over Wasserstein (WoW) distance between Q^1 and Q^2 is

$$\mathcal{W}_{\mathcal{W}}(Q^1, Q^2) = \inf_{\pi \in \Gamma(Q^1, Q^2)} \int_{\mathcal{P}(\mathbb{X})^2} \mathcal{W}(P^1, P^2)\, \pi(dP^1, dP^2).$$

Here, $\Gamma(Q^1, Q^2)$ denotes the set of probability measures on $\mathcal{P}(\mathbb{X})\times\mathcal{P}(\mathbb{X})$ with marginals Q^1 and Q^2. Again, Remark 7.1.7 in Luigi et al. (2008) and the fact that $\mathcal{W}$ has a bounded diameter imply that the metric space $(\mathcal{P}(\mathcal{P}(\mathbb{X})), \mathcal{W}_{\mathcal{W}})$ is a Polish space with a bounded diameter.

The second distance function we consider is the Hierarchical Integral Probability Metric (HIPM), introduced in Catalano and Lavenant (2024). The primary intuition behind its definition is as follows: let Q^1, Q^2 be probability measures on $\mathcal{P}(\mathbb{X})$ and let $\widetilde{P}^1 \sim Q^1, \widetilde{P}^2 \sim Q^2$. We compute the distance between Q^1 and Q^2 by considering the distances between the random integrals $\widetilde{P}^1(f)$ and $\widetilde{P}^2(f)$, where f ranges over a class of functions from $\mathbb{X}$ to $\mathbb{R}$. By considering these projections, we shift the focus from $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ to $\mathcal{P}(\mathbb{R})$, the latter of which is significantly easier to handle. As noted in Catalano and Lavenant (2024), such a projection is justified by the fact that for a random probability measure $\widetilde{P}$, the laws of the random integrals $\widetilde{P}(f)$, where f belongs to the set of continuous and bounded functions, are sufficient to characterize the law of $\widetilde{P}$. This leads to the following definition:

Definition 2.4 (Hierarchical integral probability metric)**.** Given $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, the hierarchical integral probability metric (HIPM) between Q^1 and Q^2 is defined as

$$d_{\mathcal{F}}(Q^1, Q^2) = \sup_{f \in \mathcal{F}} \mathcal{W}(\mathcal{L}(\widetilde{P}^1(f)), \mathcal{L}(\widetilde{P}^2(f))),$$

where $\widetilde{P}^k \sim Q^k$ for $k = 1, 2$ and $\mathcal{F}$ is a class of bounded and measurable functions from $\mathbb{X}$ to $\mathbb{R}$.

Alternatively, the HIPM can be expressed as

$$d_{\mathcal{F}}(Q^1, Q^2) = \sup_{f \in \mathcal{F}} \mathcal{W}(Q^1 \circ \hat{f}^{-1}, Q^2 \circ \hat{f}^{-1}),$$

where $\hat{f}$ denotes the lifting of the function f, i.e., $\hat{f}(P) = \int_{\mathbb{X}} f \, dP$, and $Q^k \circ \hat{f}^{-1}$ is the push-forward measure.

As long as the function class $\mathcal{F}$ is sufficiently rich, this construction defines a valid distance on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$. Hereafter, we set $\mathcal{F} = \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})$ and denote the corresponding metric $d_{\mathcal{F}}$ by d_{Lip}.

Both distance functions possess desirable topological properties. In particular, by Theorem 3.1 in Catalano and Lavenant (2024), WoW metrizes weak convergence on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$, and HIPM does the same when $\mathbb{X}$ is compact.

There is a relation between the two distance functions; specifically, they are special cases of an Integral Probability Metric defined on a generic Polish space.

Definition 2.5 (Integral Probability Metric)**.** Let $(\mathbb{Y}, d_{\mathbb{Y}})$ be a Polish space and $\mathcal{H}$ a class of bounded and measurable functions from $\mathbb{Y}$ to $\mathbb{R}$. The Integral Probability Metric (IPM) between $P^1, P^2 \in \mathcal{P}(\mathbb{Y})$ is

$$\mathcal{I}_{\mathcal{H}}(P^1, P^2) = \sup_{f \in \mathcal{H}} \left| \int_{\mathbb{Y}} f(y) P^1(dy) - \int_{\mathbb{Y}} f(y) P^2(dy) \right|.$$

Firstly, let us show that WoW can be written as an integral probability metric. By Remark 6.5 Villani et al. (2008), if P^1, P^2 belong to $\mathcal{P}_1(\mathbb{X})$, the Wasserstein space of order 1, the Wasserstein distance can be written as

$$\mathcal{W}(P^1, P^2) = \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \left| \int_{\mathbb{X}} f(x) P^1(dx) - \int_{\mathbb{X}} f(x) P^2(dx) \right|,$$

where $\mathrm{Lip}_1(\mathbb{X}, \mathbb{R})$ denotes the set of $1-$Lipschitz continuous functions from $\mathbb{X}$ to $\mathbb{R}$. Since $(\mathcal{P}(\mathcal{P}(\mathbb{X})), \mathcal{W}_{\mathcal{W}})$ is a Polish space with a bounded diameter, we can use the above result to obtain, for all $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$,

$$\mathcal{W}_{\mathcal{W}}(Q^1, Q^2) = \sup_{f \in \mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})} \left| \int_{\mathcal{P}(\mathbb{X})} f(P) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} f(P) Q^2(dP) \right|,$$

where $\mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})$ denotes the set of $1-$Lipschitz continuous functions from $\mathcal{P}(\mathbb{X})$ to $\mathbb{R}$ with respect to $\mathcal{W}$.

Similarly, we show that HIPM can be written as an integral probability metric. For $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$,

$$\begin{aligned}
d_{\mathrm{Lip}}(Q^1, Q^2) &= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \mathcal{W}(Q^1 \circ \hat{f}^{-1}, Q^2 \circ \hat{f}^{-1}) \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \sup_{g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})} \left| \int_{\mathbb{R}} g(x)(Q^1 \circ \hat{f}^{-1})(dx) - \int_{\mathbb{R}} g(x)(Q^2 \circ \hat{f}^{-1})(dx) \right| \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \sup_{g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})} \left| \int_{\mathcal{P}(\mathbb{X})} g(\widehat{f}(P)) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} g(\widehat{f}(P)) Q^2(dP) \right| \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \sup_{g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})} \left| \int_{\mathcal{P}(\mathbb{X})} g(P(f)) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} g(P(f)) Q^2(dP) \right| \\
&= \sup_{h \in \mathcal{D}} \left| \int_{\mathcal{P}(\mathbb{X})} h(P) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} h(P) Q^2(dP) \right|,
\end{aligned}$$

where $\mathcal{D} = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)),\ g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R}), f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})\}$.

By representing both distance functions as Integral Probability Metrics, we see that the HIPM is upper bounded by the WoW distance. Indeed, we can show that $\mathcal{D}$ is a subset of $\mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})$: let $h \in \mathcal{D}$ and $P^1, P^2 \in \mathcal{P}(\mathbb{X})$, then

$$|h(P^1) - h(P^2)| = |g(P^1(f)) - g(P^2(f))| \leq |P^1(f) - P^2(f)| \leq \sup_{f \in \mathrm{Lip}_1(\mathbb{X},\mathbb{R})} |P^1(f) - P^2(f)|$$
$$= \mathcal{W}(P^1, P^2).$$

Remark 2.6. The hierarchical estimator $Q_{n,m}$ is used to estimate the law of the random probability measure Q. The effectiveness of this estimator can be studied using either WoW or HIPM. A key difference between these distance functions lies in the rate of convergence of the expected distance between $Q_{n,m}$ and Q. In particular, by Theorem 4.1 in Catalano and Lavenant (2024), if $\mathbb{X}$ is a bounded subset of $\mathbb{R}^d$, then

$$\mathbb{E}\mathcal{W}_{\mathcal{W}}(Q_n, Q) = \mathcal{O}\left(\frac{\log\log(n)}{\log(n)}\right),$$

where $\widetilde{P}_1, \ldots, \widetilde{P}_n \overset{\text{i.i.d.}}{\sim} Q$ and $Q_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{\widetilde{P}_i}$. Note that since we only observe $Q_{n,m}$, its convergence rate will be slower than that of Q_n. Moreover, the theorem shows that if Q is a Dirichlet process, the convergence is faster than any polynomial rate.

On the other hand, using the HIPM, we obtain a faster convergence rate. In particular, by Theorem 4.3 in Catalano and Lavenant (2024), if $\mathbb{X}$ is a bounded subset of $\mathbb{R}^d$, then

$$\mathbb{E}d_{\mathrm{Lip}}(Q_{n,m}, Q) = \begin{cases} \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right), & \text{if } d = 1 \\ \mathcal{O}\left(\frac{1}{\sqrt{n}}\log(n) + \frac{1}{\sqrt{m}}\log(m)\right), & \text{if } d = 2 \\ \mathcal{O}\left(\frac{1}{n^{\frac{1}{d}}} + \frac{1}{m^{\frac{1}{d}}}\right), & \text{if } d > 2. \end{cases}$$

Such rates of convergence play an important role in studying the properties of the thresholds in Section 2.4.

To summarize, we consider the following hypothesis testing problem. Let d denote either the WoW or the HIPM distance, and let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. We test the null hypothesis

$$H_0 : d(Q^1, Q^2) = 0$$

against the alternative

$$H_1 : d(Q^1, Q^2) > 0.$$

The test is based on independent samples

$$\widehat{P}^1_{1,m}, \ldots, \widehat{P}^1_{n,m} \overset{\text{i.i.d.}}{\sim} Q^{1,(m)}, \qquad \widehat{P}^2_{1,m}, \ldots, \widehat{P}^2_{n,m} \overset{\text{i.i.d.}}{\sim} Q^{2,(m)}.$$

The testing scheme we develop will reject H_0 if the distance between the hierarchical estimators exceeds some threshold. In the following sections, we discuss methods for determining this threshold.

2.4 Theoretical thresholds via Rademacher complexity

This section develops a uniform, nonparametric threshold for hypothesis testing derived from Rademacher complexity. The primary objective is to establish a concentration inequality for the distance between two hierarchical estimators. Intuitively, given two laws of random probability measures Q^1, Q^2, whether the distance function d is WoW or HIPM, $d(Q^1_{n,m}, Q^2_{n,m})$ should be close to $d(Q^1, Q^2)$. A concentration inequality quantifies how close these quantities are for a given probability level. In particular, for $\theta \in (0,1)$, it provides a radius $r(n,m,\theta)$, depending on n, m, and θ, such that the probability that $d(Q^1_{n,m}, Q^2_{n,m})$ is within radius $r(n,m,\theta)$ of $d(Q^1, Q^2)$ is at least $1-\theta$. This implies that under the null hypothesis, for every $\theta \in (0,1)$, $d(Q^1_{n,m}, Q^2_{n,m})$ will be smaller than the $r(n,m,\theta)$ with probability at least $1-\theta$. Therefore, if we use $r(n,m,\theta)$ as a threshold function and reject H_0 whenever

$$d(Q^1_{n,m}, Q^2_{n,m}) > r(n,m,\theta),$$

we are guaranteed that the probability of making a Type I error is at most θ. Let us now discuss how to obtain the threshold function $r(n,m,\theta)$ and its relationship to the Rademacher complexity of the function class that defines the distance function.

The main tool for obtaining such a result is Talagrand's inequality (see Proposition B.1 in Sriperumbudur et al. (2012)). The radius function, as described above, is provided for a random variable of a specific form, which includes integral probability metrics. The dominant term in the radius function is the expected value of the random variable of interest. Following standard symmetrization arguments from empirical process theory, we bound this expectation using the Rademacher complexity of the underlying function class. This approach yields the following theoretical results:

Theorem 2.7 (Threshold for HIPM)**.** *Let $\mathbb{X}$ be a compact subset of $\mathbb{R}^d$, and let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Then, there exists a function $r_d(n,m,\theta)$ such that for all $n, m \geq 1$ and $\theta \in (0,1)$,*

$$\mathbb{P}\left(\left|d_{\text{Lip}}(Q^1_{n,m}, Q^2_{n,m}) - d_{\text{Lip}}(Q^1, Q^2)\right| \geq r_d(n,m,\theta)\right) \leq \theta,$$

where for a fixed θ,

$$r_d(n,m,\theta) = \begin{cases} \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right), & \text{if } d = 1 \\ \mathcal{O}\left(\frac{1}{\sqrt{n}}\log(n) + \frac{1}{\sqrt{m}}\log(m)\right), & \text{if } d = 2 \\ \mathcal{O}\left(\frac{1}{n^{\frac{1}{d}}} + \frac{1}{m^{\frac{1}{d}}}\right), & \text{if } d > 2. \end{cases}$$

A similar theorem is obtained for the Wasserstein over Wasserstein distance.

Theorem 2.8 (Threshold for WoW)**.** *Let $\mathbb{X}$ be a compact subset of $\mathbb{R}^d$, and let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Then, there exists an integer $N \geq 1$ and a function $r_d(n,m,\theta)$ such that for all $n \geq N$, $m \geq 1$, and $\theta \in (0,1)$,*

$$\mathbb{P}\left(\left|\mathcal{W}_{\mathcal{W}}(Q^1_{n,m}, Q^2_{n,m}) - \mathcal{W}_{\mathcal{W}}(Q^1, Q^2)\right| \geq r_d(n,m,\theta)\right) \leq \theta,$$

where for a fixed θ,

$$r_d(n,m,\theta) = \begin{cases} \mathcal{O}\left(\frac{\log(\log(n))}{\log(n)} + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right), & \text{if } d = 1 \\ \mathcal{O}\left(\frac{\log(\log(n))}{\log(n)} + \frac{1}{\sqrt{n}}\log(n) + \frac{1}{\sqrt{m}}\log(m)\right), & \text{if } d = 2 \\ \mathcal{O}\left(\frac{\log(\log(n))}{\log(n)} + \frac{1}{n^{\frac{1}{d}}} + \frac{1}{m^{\frac{1}{d}}}\right), & \text{if } d > 2. \end{cases}$$

Remark 2.9. If $\mathbb{X} = [a,b]$, the exact form of the threshold function in Theorem 2.7 can be derived, as the bound for the Rademacher complexity is known in closed form. In this case, we obtain a finite-sample guarantee for the Type I error. Note that for the WoW distance, even if the exact threshold were known, the finite-sample guarantee would hold only for $n \geq N$ for some integer $N \geq 1$.

Remark 2.10. Theorems 2.7 and 2.8 can be extended to more general spaces $\mathbb{X}$, provided that one can bound the Rademacher complexity of the function class over which the IPM is defined. Whether an exact threshold can be obtained depends on whether the bound on the Rademacher complexity is exact or merely an asymptotic upper bound.

The consequence of the theorems above is a testing scheme which, given a significance level θ, rejects the null hypothesis whenever the observed distance between the hierarchical estimators exceeds the threshold from the aforementioned theorems. In addition to controlling the Type I error, these theorems also ensure consistency. Intuitively, under H_1, the observed distance $d(Q^1_{n,m}, Q^2_{n,m})$, whether d is WoW or HIPM, will be close to $d(Q^1, Q^2)$. Hence, as long as the threshold $r_d(n,m,\theta)$ converges to 0 as $n, m \to \infty$, we will eventually have $d(Q^1_{n,m}, Q^2_{n,m}) > r_d(n,m,\theta)$. The theorems below formalize this intuition. Note that in the limits below, there is no dependency between n and m in how they converge to infinity.

Theorem 2.11 (Consistency for HIPM)**.** *Let $\mathbb{X}$ be a compact subset of $\mathbb{R}^d$, and let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Let θ denote the significance level, and let $r_d(n, m, \theta)$ be the threshold function from Theorem 2.7. Then:*

- *Type I error: If $Q^1 = Q^2$, then*

$$\mathbb{P}\left(d_{\mathrm{Lip}}(Q^1_{n,m}, Q^2_{n,m}) > r_d(n, m, \theta)\right) \leq \theta \quad \textit{for all } n, m \geq 1.$$

- *Consistency: If $Q^1 \neq Q^2$, then*

$$\lim_{n,m\to\infty} \mathbb{P}\left(d_{\mathrm{Lip}}(Q^1_{n,m}, Q^2_{n,m}) > r_d(n, m, \theta)\right) = 1.$$

Theorem 2.12 (Consistency for WoW)**.** *Let $\mathbb{X}$ be a compact subset of $\mathbb{R}^d$, and let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Let θ denote the significance level, and let $r_d(n, m, \theta)$ be the threshold function from Theorem 2.8. Then, there exists $N \geq 1$ such that:*

- *Type I error: If $Q^1 = Q^2$, then*

$$\mathbb{P}\left(\mathcal{W}_{\mathcal{W}}(Q^1_{n,m}, Q^2_{n,m}) > r_d(n, m, \theta)\right) \leq \theta \quad \textit{for all } n \geq N, m \geq 1.$$

- *Consistency: If $Q^1 \neq Q^2$, then*

$$\lim_{n,m\to\infty} \mathbb{P}\left(\mathcal{W}_{\mathcal{W}}(Q^1_{n,m}, Q^2_{n,m}) > r_d(n, m, \theta)\right) = 1.$$

While theoretically appealing, the aforementioned theorems have practical limitations. Achieving high power requires the threshold to decay to zero at a sufficiently fast rate. However, the logarithmic terms present in the threshold for the WoW distance result in a threshold function that converges to zero at a suboptimal rate. Furthermore, although using the HIPM provides a parametric rate, the magnitude of the constant term within the function often makes the testing scheme impractical for finite samples.

These limitations are demonstrated by our numerical simulations. We restrict our simulations to the HIPM, as the threshold associated with the WoW distance is even more conservative than that of the HIPM.

Suppose that we have two different laws of RPMs, $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. For a fixed probability level $\theta \in (0, 1)$, to obtain a low Type II error, the theoretical threshold should be less than the empirical threshold. The empirical threshold is defined as the $(1-\theta)$-quantile of the random variable $d_{\mathrm{Lip}}(Q^1_{n,m}, Q^2_{n,m})$. In the figure below, we display how the theoretical and empirical thresholds differ for a fixed θ as n and m increase.

Fix $\theta = 0.15$ and consider two Dirichlet processes $Q^1 = \mathrm{DP}(1.0, P_0)$ and $Q^2 = \mathrm{DP}(1.5, P_0)$, where $P_0 = \mathrm{Law}(Z)$ and

$$Z = \begin{cases} 0.25X - 1, & \text{with probability } 0.5, \\ 0.25X + 0.75, & \text{with probability } 0.5, \end{cases}$$

with $X \sim \mathrm{Unif}(0,1)$.

As shown in Figure 2.1, the theoretical threshold is significantly larger than the empirical threshold, even for large values of n and m. Consequently, the null hypothesis is never rejected when it is false. The reason for such a large theoretical threshold is that it is designed to be uniform over the entire space of laws of random probability measures.

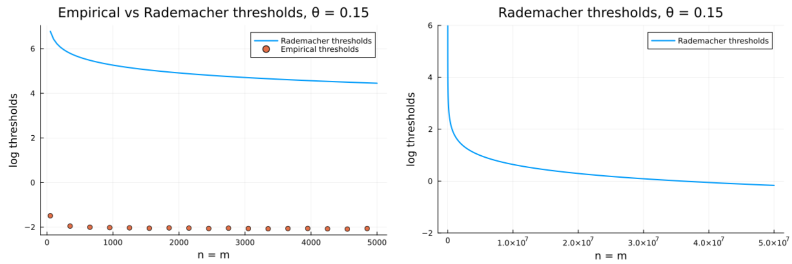


Figure 2.1

2.5 Practical thresholds via a permutation approach

In this section, we introduce a permutation-based framework for two-sample testing that yields improved performance in practice. As discussed earlier, the threshold derived from Rademacher complexity is overly conservative, resulting in low power for finite sample sizes. The permutation approach alleviates this issue by providing a more accurate approximation of the quantile function of the test statistic under the null hypothesis H_0, while typically yielding a much smaller threshold under the alternative. Recall that, given $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, we observe

$$\widehat{P}^1_{1,m}, \cdots, \widehat{P}^1_{n,m} \overset{\text{i.i.d}}{\sim} Q^{1,(m)}, \quad \widehat{P}^2_{1,m}, \cdots, \widehat{P}^2_{n,m} \overset{\text{i.i.d}}{\sim} Q^{2,(m)}.$$

Using these samples, we compute the distance $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$, where d can be either WoW or HIPM. Our goal is to estimate the quantile function of the random variable $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$ assuming that $Q^1 = Q^2$, and use it as a testing threshold. Ideally, such a quantile could be estimated empirically from multiple independent realizations of this random variable. In practice, however, only a single realization is available.

We circumvent this limitation by noting that under the null hypothesis $Q^1 = Q^2$, all random probability measures $\widehat{P}^k_{i,m}$ share the same distribution. Consequently, if we randomly shuffle these objects and recompute the distance, the resulting value constitutes another realization from the distribution of $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$. By repeating this process, we can obtain several samples of the distance between hierarchical estimators. Although these samples are not independent, this dependence is typically negligible in practice when the sample sizes n and m are sufficiently large.

We now formalize this procedure. Recall the hierarchical estimators

$$\widehat{Q}^1_{n,m} = \frac{1}{n}\sum_{i=1}^{n} \delta_{\widehat{P}^1_{i,m}} \quad \text{and} \quad \widehat{Q}^2_{n,m} = \frac{1}{n}\sum_{i=1}^{n} \delta_{\widehat{P}^2_{i,m}}.$$

To implement the permutation test, we first form the pooled collection of random probability measures

$$\left\{\widehat{P}_{1,m}, \cdots, \widehat{P}_{2n,m}\right\} := \left\{\widehat{P}^1_{1,m}, \dots, \widehat{P}^1_{n,m}, \widehat{P}^2_{1,m}, \dots, \widehat{P}^2_{n,m}\right\}.$$

A random permutation of this collection is induced by a vector $R = (R_1, \dots, R_{2n})$ sampled uniformly from the set of all permutations of $\{1, \dots, 2n\}$. For notational brevity, we omit the explicit dependence of R on n and m.

We then define the permuted hierarchical estimators as

$$\widetilde{Q}^1_{n,m} = \frac{1}{n}\sum_{i=1}^{n} \delta_{\widehat{P}_{R_i,m}} \quad \text{and} \quad \widetilde{Q}^2_{n,m} = \frac{1}{n}\sum_{i=n+1}^{2n} \delta_{\widehat{P}_{R_i,m}}.$$

Finally, we compute the distance $d(\widetilde{Q}^1_{n,m}, \widetilde{Q}^2_{n,m})$.

Repeating this process, randomly permuting the pooled sample and recomputing the distance, yields the approximate samples of the distance between the hierarchical estimators of Q^1 and Q^2. Then, for a given significance level $\theta \in (0, 1)$, we set the testing threshold equal to the empirical $(1-\theta)$-quantile of these values.

In practice, since we work with the hierarchical samples $\mathbf{X}_{n,m}, \mathbf{Y}_{n,m}$ permutation is implemented by exchanging the rows between or across them. The resulting procedure is summarized in Algorithm 1.

Algorithm 1 Permutation test with $d \in \{d_{\text{Lip}}, \mathcal{W}_{\mathcal{W}}\}$

Require: Hierarchical samples $\mathbf{X}_{n,m}, \mathbf{Y}_{n,m}$; significance level $\theta \in (0,1)$; number of permutations n_{perm}

Ensure: Decision: 1 = reject H_0, 0 = do not reject H_0

```
T_obs ← d(X_{n,m}, Y_{n,m})                        ▷ Observed test statistic
Z ← vcat(X_{n,m}, Y_{n,m})                         ▷ Pool the 2n rows
for i = 1, ..., n_perm do
    R ← permute({1, ..., 2n})
    T_i ← d(Z_{R_{1:n}}, Z_{R_{n+1:2n}})
end for
r_θ ← q̂_{1−θ}(T_1, ..., T_{n_perm})                ▷ Empirical (1 − θ)-quantile
return 1{T_obs > r_θ}
```

The theoretical properties of the proposed procedure are studied by analyzing the asymptotic distributions of the observed test statistic and the corresponding threshold under the null and alternative hypotheses. To this end, we appropriately rescale the observed distance and define the test statistic

$$T_{n,m} = \sqrt{\frac{n}{2}} d_{\text{Lip}} \left(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m} \right).$$

The corresponding threshold is defined as

$$r(n,m,\theta) = \inf \left\{ t \in \mathbb{R} : \mathbb{P}\left(\widetilde{T}_{n,m} > t \right) \leq \theta \right\},$$

where

$$\widetilde{T}_{n,m} = \sqrt{\frac{n}{2}}\, d(\widetilde{Q}^1_{n,m}, \widetilde{Q}^2_{n,m}).$$

Note that this scaling is introduced solely for theoretical analysis and is not required for the practical implementation of the algorithm.

In this section, we restrict attention to the case $d = d_{\text{Lip}}$. Deriving the asymptotic distribution of $T_{n,m}$ depends on whether the function class over which the corresponding integral probability metric is defined is a Donsker class (see the proof of Theorem **??**). At present, we are unable to verify this property for the WoW distance, and therefore do not establish the theoretical guarantees.

To conclude, we establish asymptotic level-θ control and consistency guarantees for the proposed testing scheme.

Theorem 2.13 (Consistency of permutation approach)**.** *Given $\mathbb{X} = [a,b]$, and $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Let θ denote the significance level and $r(n,m,\theta)$ be a threshold from permutation* *approach. Then*

- *if* $Q^1 = Q^2$, *then*
$$\limsup_{n,m\to\infty} \mathbb{P}\left(T_{n,m} > r(n,m,\theta)\right) \leq \theta.$$
- *if* $Q^1 \neq Q^2$, *then*
$$\lim_{n,m\to\infty} \mathbb{P}\left(T_{n,m} > r(n,m,\theta)\right) = 1.$$

2.6 Simulation studies

In this section, we examine the finite-sample performance of the proposed HIPM and WoW testing procedures. Our numerical analysis is organized into three primary components. First, we investigate the validity of the permutation-based test by evaluating the Type I error rate and statistical power of both HIPM and WoW using a synthetic dataset. Second, we compare our methods with those existing in the literature. Finally, we demonstrate the practical utility of the proposed framework through an application to a mortality dataset.

2.6.1 Empirical comparison between HIPM and WoW

We compare the use of the HIPM and WoW within our permutation-based test by analysing the Type I error, the power, and the ROC curves in several simulated settings, taking into account both parametric and nonparametric data-generating processes, both in settings that satisfy the assumptions of Theorem **??** for asymptotic guarantees, and settings that do not satisfy them, notably $\mathbb{X} = \mathbb{R}$, but are still interesting in practice. We consider the following baselines for parametric and nonparametric models.

1. Parametric model that fulfills the assumptions of Theorem **??** (PA):
$$Q^1 = \mathcal{L}(\tilde{P}), \qquad \tilde{P}|\tilde{\mu} = \mathrm{Beta}(\tilde{\mu}, 1-\tilde{\mu}), \qquad \tilde{\mu} \sim \mathrm{Beta}(a_0 = 1, b_0 = 1);$$
2. Parametric model that does not fulfill the assumptions of Theorem **??** (PNA):
$$Q^1 = \mathcal{L}(\tilde{P}), \qquad \tilde{P}|\tilde{\mu} = \mathcal{N}(\tilde{\mu}, \sigma^2 = 1), \qquad \tilde{\mu} \sim \mathcal{N}(\mu_0 = 0, \sigma_0^2 = 1);$$
3. Nonparametric model that fulfills the assumptions of Theorem **??** (NPA):
$$Q^1 = \mathrm{DP}(\alpha = 1, P_0 = \mathrm{Unif}[0,1]);$$

4. Nonparametric model that does not fulfill the assumptions of Theorem **??** (NPNA)

$$Q^1 = \mathrm{DP}(\alpha = 1, P_0 = N(0,1)),$$

where $\mathrm{Beta}(a,b)$ is the Beta distribution with shape parameters a, b, $\mathcal{N}(\mu, \sigma^2)$ is the Gaussian distribution with mean μ and variance σ^2, $\mathrm{Unif}[0,1]$ is the Uniform distribution on the interval $[0,1]$, and $\mathrm{DP}(\alpha, P_0)$ denotes the Dirichlet Process with concentration $\alpha > 0$ and base probability $P_0 \in \mathcal{P}(\mathbb{X})$ (Ferguson, 1973). Note that if P_0 has full support on $\mathbb{X}$, $\mathrm{DP}(\alpha, P_0)$ has full weak support on $\mathcal{P}(\mathbb{X})$.

To investigate the behaviour of the Type I error, we simulate the data from $Q^1 = Q^2$ equal to each of the examples above $S = 1000$ times, perform our permutation-based test for both the WoW and the HIPM, and report the percentage of times it rejects the (correct) $H_0 = Q^1 = Q^2$ for each significance level θ. In each simulation we set $n = m = n_{\text{perm}} = 100$. The results are shown in Figure **[MC: put figure]**, where we see that both WoW and HIPM recover a Type I error which is approximately equal to the significance level θ, in all simulation settings. Interestingly, the WoW distance appears to control the Type I error even if our theory does not guaranty it.

To investigate the power of the test, we simulate the data under the alternative hypothesis $Q^1 \neq Q^2$. Specifically, we take Q^1 equal to each of the settings above, and modify its parameters in two different ways for each setting: for the parametric models, we change both the 'likelihood' and the 'prior', for the Dirichlet process, we change both the concentration parameter and the base measure. We consider the following alternative hypotheses:

1a. $Q^2 = \mathcal{L}(\tilde{P})$, where $\tilde{P}|\tilde{\mu} = \mathrm{Beta}(2\tilde{\mu}, 2(1-\tilde{\mu}))$, and $\tilde{\mu} \sim \mathrm{Beta}(a_0 = 1, b_0 = 1)$;

1b. $Q^2 = \mathcal{L}(\tilde{P})$, where $\tilde{P}|\tilde{\mu} = \mathrm{Beta}(\tilde{\mu}, 1-\tilde{\mu})$, and $\tilde{\mu} \sim \mathrm{Beta}(a_0 = 2, b_0 = 1)$;

2a. $Q^2 = \mathcal{L}(\tilde{P})$, where $\tilde{P}|\tilde{\mu} = \mathcal{N}(\tilde{\mu}, \sigma^2 = 2)$, and $\tilde{\mu} \sim \mathcal{N}(\mu_0 = 0, \sigma_0^2 = 1)$;

2b. $Q^2 = \mathcal{L}(\tilde{P})$, where $\tilde{P}|\tilde{\mu} = \mathcal{N}(\tilde{\mu}, \sigma^2 = 1)$, and $\tilde{\mu} \sim \mathcal{N}(\mu_0 = 1, \sigma_0^2 = 1)$;

3a. $Q^2 = \mathrm{DP}(\alpha = 2, P_0 = \mathrm{Unif}[0,1])$;

3b. $Q^2 = \mathrm{DP}(\alpha = 1, P_0 = \mathrm{Beta}(2,2))$;

4a. $Q^2 = \mathrm{DP}(\alpha = 2, P_0 = N(0,1))$;

4b. $Q^2 = \mathrm{DP}(\alpha = 1, P_0 = N(1,1))$.

As before, we sample the observations $S = 1000$ times, perform our permutation-based test for both the WoW and the HIPM, and report the empirical power, that is, 1 minus the percentage of times it rejects the (incorrect) $H_0 = Q^1 = Q^2$ for each significance level θ, setting $n = m = n_{\text{perm}} = 100$.

[MC: Describe ROC curve and why we also add it]

[MC: Describe results carefully]

[MC: Underline if there is an empirical difference between performance of WoW and HIPM and, if not, underline the asymptotic guarantees and the computational times.]

[MC: Decide which plot/plots to put here and which to move in appendix if they all convey the same message.]

[MC: Find ways to make plots nicer, understand which simulations can be grouped in the same plot - save the data so that we can play with the plots afterwards.] [MC: Not for now but to keep in mind for the future: After looking at the results pick one relevant setting and add sensitivy analysis for n, m, n_{perm}, changing one at a time in the set $\{50, 100, 150, 200\}$. Put results in appendix and refer to them to justify your choices.]

2.6.2 Synthetic data

Let us start with the comparison of HIPM and WoW under the permutation approach. To see the behaviour of the Type I error, we fix two same Dirichlet Processes

$$Q^1 = \text{DP}(1.0, \text{Unif}(0,1)), \quad Q^2 = \text{DP}(1.0, \text{Unif}(0,1)),$$

and look at the rejection rates for each significance level θ.

For investigating the power, we fix two different Dirichlet processes

$$Q^1 = \text{DP}(1.0, \text{Unif}(0,1)), \quad Q^2 = \text{DP}(1.0, \text{Beta}(1, 1.4)).$$

We first take the small size for the hierarchical samples, $n = 50, m = 100$, and then increase to $n = 100, m = 100$. We also set $n_{\text{perm}} = 100$, which is the number of times we permute the observed samples. To estimate the rejection rates, we perform $S = 1000$ Monte-Carlo simulations.

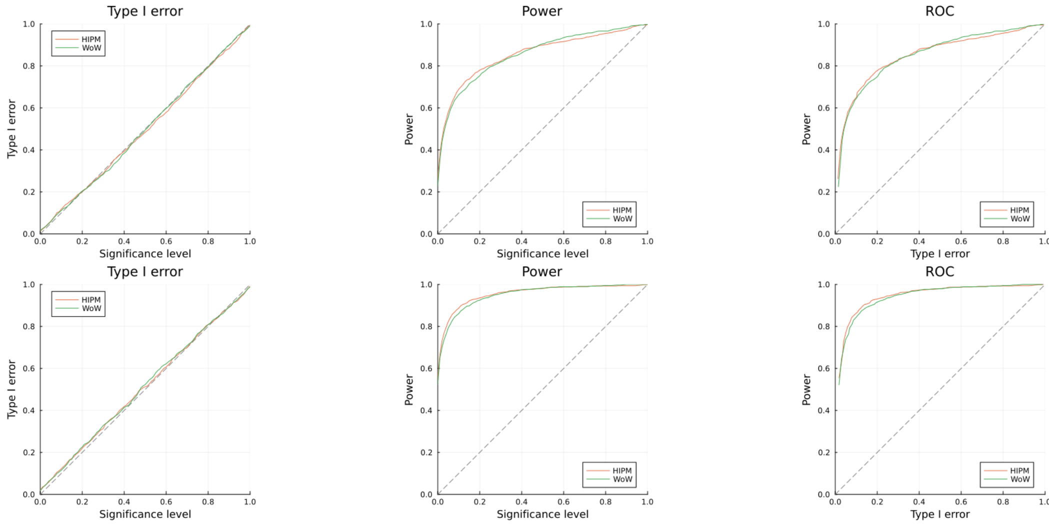


Figure 2.2

As we see from the plot, for both of the distance functions, the Type I error is well controlled. Also, Type II error is becoming low as we increase n and m. In addition, the ROC curve is much better for both distance functions than the random guessing.

What is not expected from the simulations is that we also control Type I error for the Wasserstein over Wasserstein. In theory, we could not show that under the null hypothesis, the test statistic has the same convergence properties as that with HIPM; however, in practice, it performs well.

Next, we benchmark our proposed scheme against existing methods in the literature. The first method we consider is the Energy test introduced in Székely et al. (2004). In that work, the authors develop a testing procedure for samples taking values in Euclidean space. This method can be extended to general metric spaces, provided that the underlying space is of strongly negative type.

In our simulations below, we take $\mathbb{X} = \mathbb{R}$ and consider the space of probability measures consisting of Gaussian distributions with variance equal to 1, i.e.,

$$\mathcal{P}(\mathbb{X}) = \{\mathcal{N}(\mu, 1) : \mu \in \mathbb{R}\}.$$

Equipped with the Wasserstein-1 distance, the metric space $(\mathcal{P}(\mathbb{X}), \mathcal{W})$ is of strongly negative type, since the Wasserstein distance between two Gaussian distributions with unit variance coincides with the absolute difference between their means. Hence, the Energy test is applicable in this setting.

As a second scheme, we consider the method proposed in Dubey and Müller (2019), which we refer to as the Fréchet test. We replicate Figures 1 and 2 from Dubey and Müller (2019).

To make a comparison between the testing schemes, we look at the rejection rates of each testing scheme for several pairs of laws of random probability measures (RPMs). Note that, since we work with Gaussian distributions with varying means, the law of a random probability measure is in one-to-one correspondence with the distribution of the mean parameter. In the left and right plots of Figure 2.3, we consider families of pairs of laws of RPMs on $\mathcal{P}(\mathbb{X})$, identified as the distributions on the mean parameter, defined as

$$\{\mathrm{TN}(0, 0.5^2, -10, 10), \mathrm{TN}(\delta, 0.5^2, -10, 10) : \delta \in [-1, 1]\},$$
$$\{\mathrm{TN}(0, 0.2^2, -10, 10), \mathrm{TN}(0, (0.2\tau)^2, -10, 10) : \tau \in [0, 3]\},$$

where $\mathrm{TN}(\delta, \tau, a, b)$ denotes the truncated normal distribution on the interval $[a, b]$ with mean δ and variance τ.

Note that the parameters $\delta = 0$ and $\tau = 1$ in the first and second families, respectively, correspond to estimating the Type I error, while other parameter values correspond to estimating the power. The other simulation parameters we use are $S = 1000$ and $n_{\text{perm}} = 100$. For the Fréchet test, we use a bootstrap approach where $n_{\text{bootstrap}} = 100$.

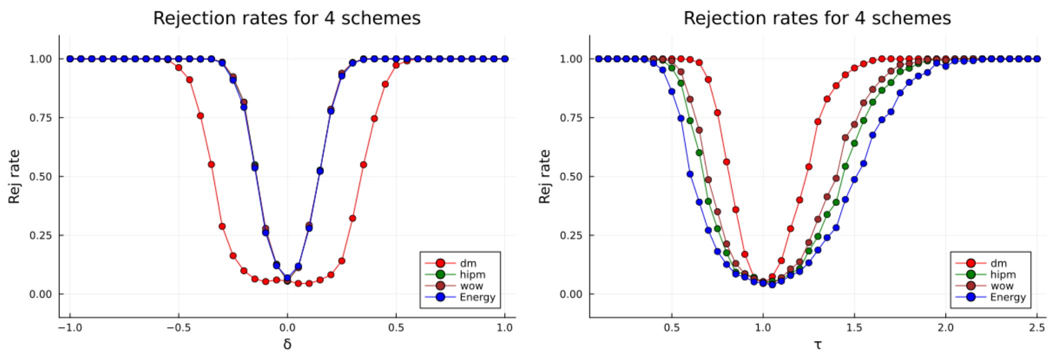


Figure 2.3

All testing schemes exhibit low Type I error. When varying the mean parameter δ in the left plot, the Energy, WoW, and HIPM tests outperform the Fréchet test. The intuition behind this behavior lies in the non-degeneracy property of distance-based tests. In particular, such tests detect the statistical summaries that are sufficient to characterize the underlying distribution; in the case of differing laws of RPMs, they capture richer distributional information than the Fréchet mean and variance alone. However, this intuition is not reflected in the right plot, where the method proposed in Dubey and Müller (2019) outperforms all other testing schemes. The final simulation on the synthetic dataset is designed to demonstrate the necessity of distance-based testing. Note that the underlying assumption behind the Fréchet test is that the Fréchet mean and variance are sufficient to distinguish between two different laws of random probability measures (RPMs). However, laws of RPMs are not always uniquely characterized by their Fréchet means and variances.

Indeed, we can construct a simple counterexample. Consider

$$
\begin{aligned}
Q^1 &= \mathcal{L}\left(\widetilde{P}^1\right) \quad \text{where} \quad \widetilde{P}^1 = \mathcal{N}\left(\widetilde{\mu}^1, 1\right), \quad \mathcal{L}\left(\widetilde{\mu}^1\right) = \tfrac{1}{2}\delta_{-1} + \tfrac{1}{2}\delta_1, \\
Q^2 &= \mathcal{L}\left(\widetilde{P}^2\right) \quad \text{where} \quad \widetilde{P}^2 = \mathcal{N}\left(\widetilde{\mu}^2, 1\right), \quad \mathcal{L}\left(\widetilde{\mu}^2\right) = \tfrac{1}{8}\delta_{-2} + \tfrac{3}{4}\delta_0 + \tfrac{1}{8}\delta_2,
\end{aligned}
$$

where $Q^1 \neq Q^2$, yet their associated Fréchet means and variances coincide (see Remarks **??**,**??**).

Moreover, if we compare Q^1 with a mixture of Q^1 and Q^2, the resulting laws remain different while still inducing the same Fréchet means and variances. Specifically, letting

$$M_\lambda = \lambda Q^1 + (1-\lambda) Q^2,$$

we consider the family of pairs

$$\{(Q^1, M_\lambda) : \lambda \in [0,1]\}.$$

Note that $\lambda = 1$ corresponds to estimating the Type I error, while other values of λ correspond to estimating the power of the testing procedures. We consider mixtures of the two laws because, as the parameter varies, it becomes more challenging for the testing schemes to detect the presence of different laws. We plot the rejection rates in Figure 2.4, where the simulation parameters are the same as those used in Figure 2.3.

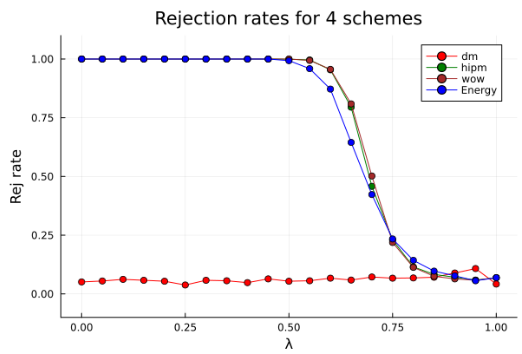


Figure 2.4

As observed in the Figure 2.4, since the two distributions induce the same Fréchet means and variances, the method proposed in Dubey and Müller (2019) fails to detect the presence of different laws. In contrast, the HIPM and WoW tests perform slightly better than Energy test in identifying the discrepancy.

To conclude, although the Fréchet test can exhibit higher power than the other testing schemes in certain settings, its main limitation is that it fails to detect the difference when the laws of RPMs are not fully characterized by the Fréchet mean and variance.

The Energy test, as well as permutation tests based on HIPM or WoW, mitigate this limitation. Moreover, HIPM and WoW may be preferable to the Energy test in practice, since the underlying metric space—in our case, the space of probability measures—may fail to be of negative type, or this property may be difficult to verify.

2.6.3 Mortality data

We apply our proposed testing scheme to human mortality data available from the Human Mortality Database (https://www.mortality.org/). The dataset contains period life tables for various countries and years. A period life table is constructed by estimating age-specific mortality rates from deaths and exposures in a given year t, which are then used to derive the age-at-death distribution for a hypothetical cohort.

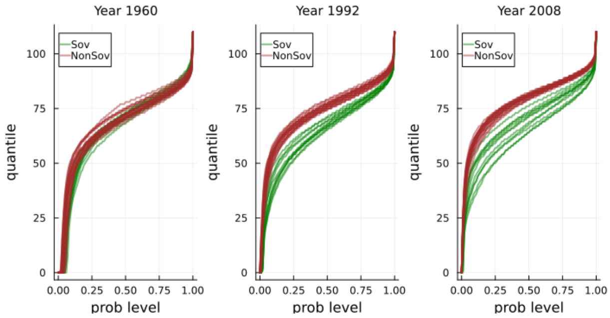
Our analysis focuses on how mortality rates differed between 1960 and 2010 when comparing countries that were under Soviet influence during the 20th century against those under Western influence. To this end, we categorize the countries into two groups, following the classification established in Dubey and Müller (2019):

- Group 1 (Soviet Influence): Belarus, Bulgaria, Czechia, Estonia, Hungary, Latvia, Lithuania, Poland, Russia, Slovakia, and Ukraine.

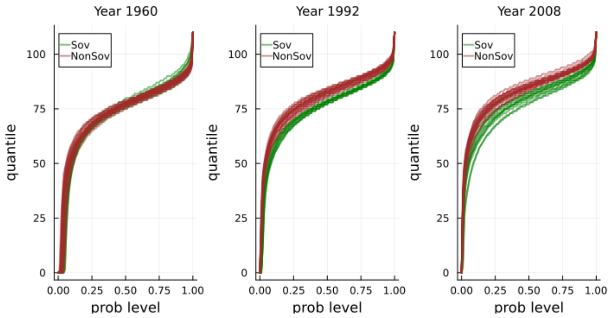
- Group 2 (Western Influence): Australia, Austria, Belgium, Canada, Denmark, Finland, France, Iceland, Ireland, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Spain, Sweden, Switzerland, United Kingdom, and United States of America.

The age range is $\{0, 1, \cdots, 110\}$, where 110 denotes that death occurred at that age or later. We choose not to truncate the age range for the following reasons. First, visual inspection of the probability mass functions and quantiles does not reveal any abnormalities, particularly at higher ages (see Figure 2.5). Second, truncating the age range leads to significant data loss. For example, among females, the maximum percentage of data lost across countries is at least 20 percent for all time periods when the age interval is truncated to $\{0, 1, \ldots, 85\}$.

In Figure 2.5, we present the quantile plots of the probability mass functions corresponding to age at death. We consider the years 1960, 1992, and 2008 to capture different historical periods: the Cold war era that is closer to the post-war period, the years shortly after the collapse of Soviet Union, and a more recent period after the breakup. The plots are shown separately for females and males. As observed, noticeable differences can be visually identified, particularly in the more recent time periods.



(a) Males



(b) Females

Figure 2.5: Quantiles for mortality rates

To assess whether there is a difference in mortality rates between the two groups of countries in each time period, we apply our proposed test separately for males and females. We set the significance level to $\theta = 0.05$ and use $n_{\mathrm{perm}} = 100$ permutations. The p-values are computed for each year.

We also compare the p-values obtained from our approach with those derived from the following alternative methods:

- Averaging: Given $\widehat{P}_1^1, \cdots, \widehat{P}_{n_1}^1$ from group 1 and $\widehat{P}_1^2, \cdots, \widehat{P}_{n_2}^2$ from group 2, we obtain average probability measures inside each group, i.e.

$$\overline{P}^1 = \frac{1}{n_1}\sum_{i=1}^{n_1} \widehat{P}_i^1, \quad \overline{P}^2 = \frac{1}{n_2}\sum_{i=1}^{n_2} \widehat{P}_i^2.$$

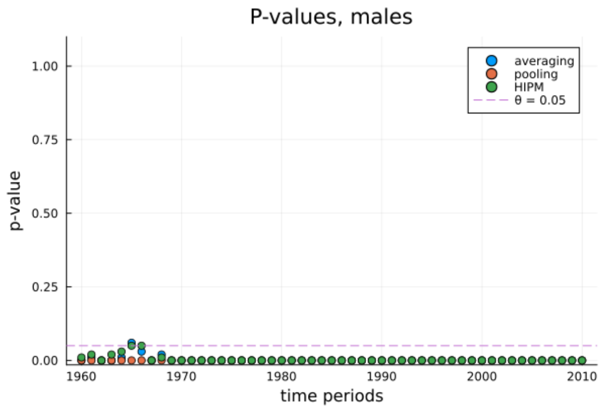
  The test statistic is obtained by computing Wasserstein-1 distance between $\overline{P}^1$ and $\overline{P}^2$. The threshold is obtained via a permutation approach, in which the observed probability measures are permuted across the two groups.

- Pooling: Within each group, we first pool the death counts across all countries. This yields a single probability measure for each group, and the test statistic is then computed as the Wasserstein-1 distance between these two measures. As before, the significance threshold is obtained via a permutation approach. The permutation procedure is implemented by reconstructing observations of ages at death from the aggregated counts in each group. Specifically, if there are k deaths at age a, we generate k observations equal to a. This is done for each group, after which the resulting observations are permuted across groups.

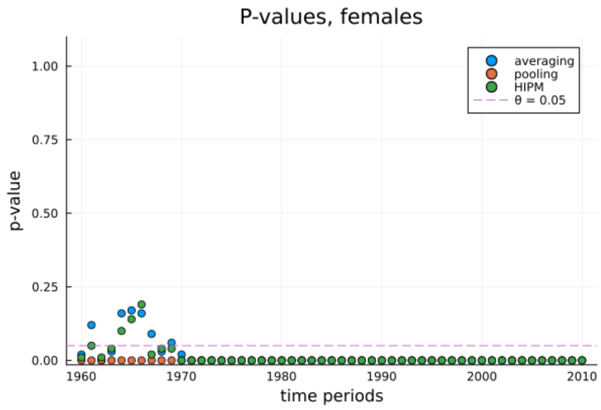
Note that the test statistic is identical for both the pooling and averaging approaches; the difference between the two testing schemes lies in the permutation procedure. In particular, the averaging approach preserves the probability measure structure within each group, whereas the pooling approach does not.

Examining the p-values obtained from the proposed method in Figure 2.6, we observe that for females, the difference between the groups is statistically significant in 1960. During the following decade, this difference becomes less significant, which may be explained by economic growth. After the 1970s, the disparity in mortality rates between the groups increases again, potentially reflecting the economic decline in the countries influenced by Soviet Union. Following the breakup of the Soviet Union, the observed differences may be associated with countries that were unable to integrate well into the Western blocks.

A similar trend is observed for males; however, the key difference is that during the period from 1960 to 1970, althouth the differences in mortality rates was decreasing, the p-values did not exceed the 0.05 threshold.



(a) Males



(b) Females

Figure 2.6: P-values for mortality rates

Note also that p-values for averaging and pooling look similar. This is because the average measure in this case is sufficient to detect the difference between two groups.

It is important to note that the pooling approach may fail to control Type I error in hierarchical settings. This limitation arises because pooling observations ignores the presence of latent random probability measures. Under hierarchical sampling, we observe n sequences, each of length m, where each sequence is generated from a different latent random probability measure. The empirical distribution obtained by pooling all observations converges to the mean measure of the law of RPM, so the test statistic converges to the distance between the mean measures of the two groups. Observations from the permutation approach are similar to samples from the mean measure; however, we obtain nm samples. Intuitively, this makes the permutation threshold converge to 0 faster than the observed distance. Note that this is not an issue if latent probability measures are identical.

Still, both of the approaches fail to detect the different laws of random probability measures if the mean measures are the same. This again underlines the importance of working with the objects that uniquely characterize the laws we want to distinguish.

Chapter 3

Merging Rate of Opinions

3.1 Introduction to Merging

In Bayesian statistics, learning about the data-generating mechanism begins with the specification of a prior distribution over a class of probability distributions that could have generated the data. This prior may be uninformative (e.g., uniform over a suitable class), reflecting limited prior information, or more concentrated when substantive prior knowledge is available. Upon observing data, the statistician updates these beliefs via Bayes' rule, obtaining a posterior distribution over the candidate data-generating laws.

The goal of this section is to compare posterior beliefs arising from different prior distributions. This problem is commonly referred to as the merging of opinions. To study it rigorously, several modeling choices must be clarified. In particular, what assumptions are made about the sequence of observations? Are restrictions imposed on the class of prior distributions?

This problem was studied in Blackwell and Dubins (1962), where the convergence of predictive distributions is analyzed. Their framework considers probability measures defined on the space of infinite sequences of observations. After observing the first n observations, one forms a predictive distribution over future observations via conditional probabilities. They show that if one probability measure Q is absolutely continuous with respect to another probability measure P, then their corresponding predictive distributions merge almost surely with respect to Q. This result applies to Bayesian settings in which priors are defined over infinite-dimensional spaces and implies that the corresponding predictive distributions, and hence posterior beliefs, merge. However, it relies on a key restriction: one probability distribution on the sequence of infinite observations induced by a prior must be absolutely continuous with respect to the other. A sufficient condition guaranteeing this is to impose absolute continuity at the level of prior distributions. However, this condition can fail even in standard Bayesian nonparametric models; for example, two Dirichlet process priors with mutually singular base measures may themselves be mutually singular. Moreover, the result is formulated under the assumption that the observations are generated according to one of the probability measures under consideration. If the objective is instead to compare posterior updating procedures themselves, it is natural to avoid strong assumptions about the true data-generating mechanism.

This perspective is adopted in Catalano and Lavenant (2023), and the goal of this chapter is to extend the study of merging to another example of a Bayesian nonparametric model.

3.2 Known Results on Merging Rates

In this section, we summarize known results on merging between two posterior distributions arising from the Bayesian model 1.1 when different priors are used. As discussed earlier, we investigate merging independently of the data-generating mechanism. That is, for a fixed sequence of observations, we compare how the corresponding posterior distributions differ under different prior specifications.

Since we work with distances between random measures, the key step is to quantify the distance between posterior distributions. To this end, we consider two priors given by completely random measures, derive the corresponding posterior distributions, and then compare them. Due to the identifiability issue discussed in Subsection 1.2.3, we compute distances between scaled versions of these posterior distributions.

We start with the case of merging between two Dirichlet processes. Recall that the Dirichlet process is obtained by normalizing a Gamma CRM. By Theorem 15 in Catalano and Lavenant (2023), for two Gamma CRMs $\tilde{\mu}^1$ and $\tilde{\mu}^2$ with parameters $(\alpha_i, 1, P_0^i)$, $i = 1, 2$, and for any sequence of observations $(x_n)_{n\geq 1}$ in $\mathbb{X}$, it holds that

$$\mathrm{AD}(\nu_S^{1,*}, \nu_S^{2,*}) \asymp \frac{1}{n}, \qquad \mathcal{W}_{\mathrm{BL}}\big(\mathcal{L}(\tilde{\mu}_S^{1,*}), \mathcal{L}(\tilde{\mu}_S^{2,*})\big) \lesssim \frac{1}{n},$$

where $\tilde{\mu}_S^{k,*}$ denotes the scaled version of the posterior CRM associated with $\tilde{\mu}^k$ and $\nu_S^{k,*}$ denotes the associated Lévy intensity for $k = 1, 2$. It is important to note that the Lévy intensities associated with scaled posterior CRMs are not random, since the posterior CRMs are no longer Cox CRMs after scaling.

The result says that the posterior distributions always merge at rate $1/n$, regardless of the observed data and independently of the data-generating mechanism.

This behavior admits an intuitive explanation. The posterior distribution induced by a Dirichlet process converges weakly to the corresponding predictive distribution, which in turn converges to the empirical measure. Equivalently, one can say that the posterior distribution concentrates around the empirical measure as the sample size increases. Since this limiting object does not depend on the prior, different posterior distributions merge.

We now discuss another known result on merging between the Dirichlet process and the generalized Gamma CRM.

Definition 3.1. A completely random measure is said to be a generalized Gamma CRM if its Lévy intensity is given by

$$d\nu(s,x) = \frac{\alpha}{\Gamma(1-\sigma)} \frac{e^{-s}}{s^{1+\sigma}} \,\mathrm{d}s \,\mathrm{d}P_0(x),$$

where $\alpha > 0$ is the total mass parameter, $\sigma \in (0,1)$ is the discount parameter, and $P_0 \in \mathcal{P}(\mathbb{X})$ is the base probability measure.

By Theorem 19 in Catalano and Lavenant (2023), let $\tilde{\mu}^1$ be a generalized Gamma CRM with parameters (α, σ, P_0) and let $\tilde{\mu}^2$ be a Gamma CRM with parameters $(\alpha, 1, P_0)$. Then,

$$\mathcal{W}_{\mathrm{AD}}\big(\mathcal{L}(\tilde{\nu}_S^{1,*}), \mathcal{L}(\tilde{\nu}_S^{2,*})\big) \asymp \max\left\{n^{-\frac{1}{1+\sigma}}, \frac{k_n}{n}\right\},$$

and

$$\mathcal{W}_{\mathrm{BL}}\big(\mathcal{L}(\tilde{\mu}_S^{1,*}), \mathcal{L}(\tilde{\mu}_S^{2,*})\big) \lesssim \max\left\{n^{-\frac{1}{1+\sigma}}, \frac{k_n}{n}\right\},$$

where k_n denotes the number of distinct observations among $x_1, \ldots, x_n$.

This result shows that merging occurs provided that the number of distinct observations does not grow at the same rate as the sample size. In such cases, the rate of merging depends on the parameter σ. In contrast to the Dirichlet process versus Dirichlet process case, merging may fail when $k_n \sim n$. This situation can arise, for instance, when the data are generated i.i.d. from a continuous distribution.

3.3 Merging between the Pitman–Yor and Dirichlet Processes

We now present a result on merging between the Pitman–Yor and Dirichlet processes. We first recall the posterior distribution derived from the Pitman–Yor process. The strategy for establishing the merging rate is to first obtain the scaled version of the posterior distributions from each completely random measure, and then study the adapted Wasserstein discrepancy.

Let $\tilde{\mu}$ be a Cox CRM with random Lévy intensity $\tilde{\nu}$, as in Definition 1.15. The random Lévy intensity has the form

$$\mathrm{d}\tilde{\nu}_\xi(s,x) = \frac{\xi\sigma}{\Gamma(1-\sigma)} \frac{e^{-s}}{s^{1+\sigma}} \mathrm{d}s \,\mathrm{d}P_0(x),$$

where $\xi \sim \Gamma\left(\frac{\theta}{\sigma}, 1\right)$.

The posterior distribution of $\tilde{\mu}$ under the Bayesian model 1.1 has the following form (Hjort et al., 2010, p. 114):

Given observations $x_1, \ldots, x_n$ from model 1.1, the posterior $\widetilde{\mu}^*$ of $\tilde{\mu}$ is a Cox CRM characterized by

$$\widetilde{\mu}^* \mid U \stackrel{d}{=} \tilde{\mu} + \sum_{i=1}^{k} J_i^U \delta_{x_i^*},$$

where

- $\tilde{\mu}$ is a generalized Gamma CRM with Lévy intensity

$$\mathrm{d}\tilde{\nu}(s, x) = \frac{\sigma}{\Gamma(1-\sigma)} \frac{e^{-Us}}{s^{1+\sigma}} \mathrm{d}s \, \mathrm{d}P_0(x).$$

  Note that we have omitted the subscript U for brevity.

- $J_i^U \sim \mathrm{Gamma}(n_i - \sigma, U)$

- $f_U(u) = \frac{\sigma}{\Gamma\left(k + \frac{\theta}{\sigma}\right)} u^{\theta + k\sigma - 1} e^{-u^\sigma} \mathbf{1}_{(0,\infty)}(u)$

and $(x_1^*, \ldots, x_k^*)$ are the distinct atoms with multiplicities $n_1, \ldots, n_k$.

Note that both k and U implicitly depend on n and are thus sequences; however, since we do not need to track this dependence explicitly before deriving the merging rate, we drop the subscript n for brevity.

Proposition 3.2. *Let $\tilde{\mu}$ be a Cox CRM associated with the Pitman–Yor process with parameters (σ, θ, P_0), and let $\widetilde{\mu}^*$ denote its posterior. Then the scaled version of $\widetilde{\mu}^*$ has random Lévy intensity*

$$\mathrm{d}\tilde{\nu}_S^*(s, x) = \mathrm{d}\rho_x^*(s) \, \mathrm{d}P_0^*(x),$$

where

$$\begin{aligned}
P_0^* &= \frac{\sigma U^\sigma}{C} P_0 + \frac{1}{C} \sum_{i=1}^{k} (n_i - \sigma) \delta_{x_i^*}, \\
\mathrm{d}\rho_x^*(s) &= \frac{C^{1-\sigma}}{\Gamma(1-\sigma)} \frac{e^{-Cs}}{s^{1+\sigma}} \mathbf{1}_{\mathbb{X} \setminus \{x_1^*, \ldots, x_k^*\}}(x) \, \mathrm{d}s + C \frac{e^{-Cs}}{s} \mathbf{1}_{\{x_1^*, \ldots, x_k^*\}}(x) \, \mathrm{d}s, \\
C &= C(\sigma, n, k, U) = \sigma U^\sigma + n - k\sigma,
\end{aligned}$$

and U is as defined above.

Proof. Since $\widetilde{\mu}^* \mid U$ is a sum of CRMs, we compute its Lévy intensity.

By iteratively applying Lemma SM4 in Catalano and Lavenant (2023), for a general CRM $\tilde{\mu}$ with Lévy intensity ν, independent and infinitely divisible random variables $J_1, \dots, J_k$ with Lévy measures π_i, and fixed atoms $y_1, \dots, y_k$, the Lévy intensity of $\tilde{\mu} + \sum_{i=1}^k J_i \delta_{y_i}$ is

$$\mathrm{d}\nu(s,x) + \sum_{i=1}^k \mathrm{d}\left(\pi_i \otimes \delta_{y_i}\right)(s,x),$$

where ν is the Lévy intensity of $\tilde{\mu}$.

The Lévy measure of $J_i^U \sim \mathrm{Gamma}(n_i - \sigma, U)$ is

$$\mathrm{d}\pi_i(s) = \frac{n_i - \sigma}{s} e^{-Us} \mathbf{1}_{(0,\infty)}(s)\,\mathrm{d}s.$$

Then, for $(s,x) \in (0,\infty) \times \mathbb{X}$,

$$\begin{aligned}
\mathrm{d}\nu(s,x) + &\sum_{i=1}^k \mathrm{d}\left(\pi_i \otimes \delta_{x_i^*}\right)(s,x) \\
&= \frac{\sigma}{\Gamma(1-\sigma)} \frac{e^{-Us}}{s^{1+\sigma}} \mathrm{d}s\,\mathrm{d}P_0(x) + \sum_{i=1}^k \frac{n_i - \sigma}{s} e^{-Us}\,\mathrm{d}s\,\mathrm{d}\delta_{x_i^*}(x) \\
&= \frac{\sigma}{\Gamma(1-\sigma)} \frac{e^{-Us}}{s^{1+\sigma}} \mathrm{d}s\,\mathrm{d}P_0(x) + \frac{e^{-Us}}{s}\,\mathrm{d}s \sum_{i=1}^k (n_i - \sigma)\,\mathrm{d}\delta_{x_i^*}(x) =: \mathrm{d}\tilde{\nu}^*(s,x).
\end{aligned}$$

So $\widetilde{\mu}^*$ is a Cox CRM with random Lévy intensity $\tilde{\nu}^*$. Denote

$$\begin{aligned}
\mathrm{d}\rho^1(s) &= \frac{\sigma}{\Gamma(1-\sigma)} \frac{e^{-Us}}{s^{1+\sigma}}\,\mathrm{d}s, \\
\mathrm{d}\rho^2(s) &= \frac{e^{-Us}}{s}\,\mathrm{d}s.
\end{aligned}$$

Then,

$$\mathrm{d}\tilde{\nu}^*(s,x) = \mathrm{d}\rho^1(s)\,\mathrm{d}P_0(x) + \mathrm{d}\rho^2(s) \sum_{i=1}^k (n_i - \sigma)\,\mathrm{d}\delta_{x_i^*}(x).$$

The next step is to obtain the scaled version of $\widetilde{\mu}^*$. By Definition 1.10, we obtain the scaled version of the latent CRM for fixed U.

As discussed in Subsection 1.2.3, if ν is the Lévy intensity of a CRM $\tilde{\mu}$, then the Lévy intensity of its scaled version is

$$\mathrm{d}\nu_S(s,x) = \mathbb{E}(\tilde{\mu}(\mathbb{X}))\,\rho_x(\mathbb{E}(\tilde{\mu}(\mathbb{X}))\,s)\,\mathrm{d}P_0(x),$$

by Lemma SM3 in Catalano and Lavenant (2023). By applying the same argument to

$$\mathrm{d}\tilde{\nu}^*(s,x) = \mathrm{d}\rho^1(s)\,\mathrm{d}P_0^1(x) + \mathrm{d}\rho^2(s)\,\mathrm{d}P_0^2(x),$$

we get

$$\mathrm{d}\tilde{\nu}_S^*(s,x) = \mathbb{E}(\widetilde{\mu}^*(\mathbb{X}) \mid U)\, \rho_x^1(\mathbb{E}(\widetilde{\mu}^*(\mathbb{X}) \mid U)\, s)\, \mathrm{d}P_0^1(x) + \mathbb{E}(\widetilde{\mu}^*(\mathbb{X}) \mid U)\, \rho_x^2(\mathbb{E}(\widetilde{\mu}^*(\mathbb{X}) \mid U)\, s)\, \mathrm{d}P_0^2(x).$$

We therefore first compute $\mathbb{E}(\widetilde{\mu}^*(\mathbb{X}) \mid U)$. By Campbell's theorem,

$$\mathbb{E}(\widetilde{\mu}^*(\mathbb{X}) \mid U) = \int_{(0,\infty)\times\mathbb{X}} s\, \mathrm{d}\tilde{\nu}^*(s,x).$$

So,

$$\begin{aligned}
\mathbb{E}(\widetilde{\mu}^*(\mathbb{X}) \mid U) &= \int_{(0,\infty)\times\mathbb{X}} s\frac{\sigma}{\Gamma(1-\sigma)}\frac{e^{-Us}}{s^{1+\sigma}}\mathrm{d}s\, \mathrm{d}P_0(x) + \int_{(0,\infty)\times\mathbb{X}} s\frac{e^{-Us}}{s}\, \mathrm{d}s \sum_{i=1}^k (n_i-\sigma)\, \mathrm{d}\delta_{x_i^*}(x) \\
&= \frac{\sigma}{\Gamma(1-\sigma)}\int_0^\infty s^{-\sigma}e^{-Us}\, \mathrm{d}s + (n-k\sigma)\int_0^\infty e^{-Us}\, \mathrm{d}s \\
&= \frac{\sigma}{\Gamma(1-\sigma)}\frac{\Gamma(1-\sigma)}{U^{1-\sigma}} + \frac{n-k\sigma}{U} \\
&= \frac{\sigma U^\sigma}{U} + \frac{n-k\sigma}{U} = \frac{C}{U},
\end{aligned}$$

where $C = C(\sigma, n, k, U) = \sigma U^\sigma + n - k\sigma$.

Plugging this into the scaling formula, we obtain the scaled Lévy intensity of the latent CRM:

$$\begin{aligned}
\mathrm{d}\tilde{\nu}_S^*(s,x) &= \frac{C}{U}\, \mathrm{d}\rho^1\left(\frac{C}{U}s\right) \mathrm{d}P_0(x) + \frac{C}{U}\, \mathrm{d}\rho^2\left(\frac{C}{U}s\right) \sum_{i=1}^k (n_i-\sigma)\, \mathrm{d}\delta_{x_i^*}(x) \\
&= \frac{C}{U}\frac{\sigma}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{C^{1+\sigma}s^{1+\sigma}}U^{1+\sigma}\, \mathrm{d}s\, \mathrm{d}P_0(x) + \frac{e^{-Cs}}{s}\, \mathrm{d}s \sum_{i=1}^k (n_i-\sigma)\, \mathrm{d}\delta_{x_i^*}(x) \\
&= \frac{\sigma}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\left(\frac{U}{C}\right)^\sigma \mathrm{d}s\, \mathrm{d}P_0(x) + \frac{e^{-Cs}}{s}\, \mathrm{d}s \sum_{i=1}^k (n_i-\sigma)\, \mathrm{d}\delta_{x_i^*}(x).
\end{aligned}$$

The above expression is not yet in canonical form; we wish to express it as a product of a transition kernel for jumps and a density for atoms.

Note that

$$\mathcal{M}_1(\tilde{\nu}_S^*) = \int_{(0,\infty)\times\mathbb{X}} s\, \mathrm{d}\tilde{\nu}_S^*(s,x) = 1,$$

since it is the Lévy intensity of a scaled CRM. The measure $s\, \mathrm{d}\tilde{\nu}_S^*$ can therefore be disintegrated as

$$s\, \mathrm{d}\tilde{\nu}_S^*(s,x) = \mathrm{d}\varrho_x(s)\, \mathrm{d}Q(x),$$

where Q is the second marginal of $s\, \mathrm{d}\tilde{\nu}_S^*$ and ϱ_x is the probability transition kernel.

Since

$$Q(\mathbb{X}) = \int_{(0,\infty)\times\mathbb{X}} s\,\mathrm{d}\tilde{\nu}_S^*(s,x) = 1,$$

Q is a probability measure. It follows that

$$\mathrm{d}\tilde{\nu}_S^*(s,x) = \frac{1}{s}\,\mathrm{d}\varrho_x(s)\,\mathrm{d}Q(x).$$

To find the terms in this decomposition, we first determine Q. For $A \in \mathcal{B}(\mathbb{X})$,

$$\begin{aligned}
Q(A) &= \int_{(0,\infty)\times A} s\,\mathrm{d}\tilde{\nu}_S^*(s,x) \\
&= \frac{\sigma}{\Gamma(1-\sigma)}\left(\frac{U}{C}\right)^{\sigma} P_0(A)\int_0^\infty s^{-\sigma}e^{-Cs}\,\mathrm{d}s + \sum_{i=1}^k (n_i-\sigma)\delta_{x_i^*}(A)\int_0^\infty e^{-Cs}\,\mathrm{d}s \\
&= \frac{\sigma}{\Gamma(1-\sigma)}\left(\frac{U}{C}\right)^{\sigma} P_0(A)\frac{\Gamma(1-\sigma)}{C^{1-\sigma}} + \frac{1}{C}\sum_{i=1}^k (n_i-\sigma)\delta_{x_i^*}(A) \\
&= \frac{\sigma U^\sigma}{C}P_0(A) + \frac{1}{C}\sum_{i=1}^k (n_i-\sigma)\delta_{x_i^*}(A).
\end{aligned}$$

We denote

$$P_0^*(\cdot) = \frac{\sigma U^\sigma}{C}P_0(\cdot) + \frac{1}{C}\sum_{i=1}^k (n_i-\sigma)\delta_{x_i^*}(\cdot).$$

It remains to determine ϱ_x, which must satisfy

$$\frac{\sigma}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{\sigma+1}}\left(\frac{U}{C}\right)^{\sigma}\mathrm{d}s\,\mathrm{d}P_0(x) + \frac{e^{-Cs}}{s}\,\mathrm{d}s\sum_{i=1}^k (n_i-\sigma)\,\mathrm{d}\delta_{x_i^*}(x) = \frac{1}{s}\,\mathrm{d}\varrho_x(s)\,\mathrm{d}P_0^*(x).$$

Since P_0 is non-atomic, we obtain

$$\mathrm{d}\varrho_x(s) = \frac{C^{1-\sigma}}{\Gamma(1-\sigma)}s^{-\sigma}e^{-Cs}\mathbf{1}_{\mathbb{X}\setminus\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s + Ce^{-Cs}\mathbf{1}_{\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s.$$

The conclusion follows by setting

$$\mathrm{d}\rho_x^*(s) = \frac{1}{s}\mathrm{d}\varrho_x(s) = \frac{C^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\mathbf{1}_{\mathbb{X}\setminus\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s + C\frac{e^{-Cs}}{s}\mathbf{1}_{\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s.$$

□

We are now ready to study the merging between the Pitman–Yor and Dirichlet processes.

Proposition 3.3. *Let $\tilde{\mu}^1$ be a Pitman–Yor CRM with parameters (σ,θ,P_0) and let $\tilde{\mu}^2$ be* *a Gamma CRM with parameters* $(\alpha, 1, P_0)$. *Then*

$$\mathcal{W}_{\mathrm{AD}}\left(\mathcal{L}\left(\tilde{\nu}_S^{*,1}\right), \mathcal{L}\left(\tilde{\nu}_S^{*,2}\right)\right) = \mathbb{E}_U\left(\mathcal{A}^U + J_1^U + J_2^U\right),$$

where

$$\begin{aligned}
\mathcal{A}^U &= \mathcal{W}_{d_{\mathbb{X},t}}\left(\frac{\sigma U^\sigma}{C} P_0 + \frac{1}{C}\sum_{i=1}^{k_n}(n_i - \sigma)\delta_{x_i^*}, \frac{\alpha}{\alpha+n} P_0 + \frac{1}{\alpha+n}\sum_{i=1}^{n}\delta_{x_i}\right),\\
J_1^U &= \frac{\sigma U^\sigma}{C}\mathcal{W}_*\left(\frac{C^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\,\mathrm{d}s, (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\right),\\
J_2^U &= \frac{n-k\sigma}{C}\mathcal{W}_*\left(\frac{Ce^{-Cs}}{s}\,\mathrm{d}s, (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\right),
\end{aligned}$$

where U *and* C *are as defined in Proposition 3.2.*

Proof. Denote by $\tilde{\nu}_S^{*,2}$ the Lévy intensity of the scaled Gamma posterior; its derivation can be found in the proof of Lemma 12 in Catalano and Lavenant (2023), and it has the following form:

$$\mathrm{d}\tilde{\nu}_S^{*,2}(s,x) = (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\left(\frac{\alpha}{\alpha+n}\,\mathrm{d}P_0(x) + \frac{1}{\alpha+n}\sum_{i=1}^{n}\mathrm{d}\delta_{x_i}(x)\right).$$

Note that the scaled Gamma posterior is no longer a Cox CRM. Since it is homogeneous, Remark 1.24 gives

$$\begin{aligned}
\mathcal{W}_{\mathrm{AD}}\left(\mathcal{L}\left(\tilde{\nu}_S^{*,1}\right), \mathcal{L}\left(\tilde{\nu}_S^{*,2}\right)\right) &= \mathbb{E}_{U\sim f_U}\left[\mathrm{AD}\left(\tilde{\nu}_S^{*,1}, \tilde{\nu}_S^{*,2}\right)\right]\\
&= \mathbb{E}_{U\sim f_U}\mathcal{W}_{d_{\mathbb{X},t}}\left(P_0^{*,1}, P_0^{*,2}\right) + \mathbb{E}_{U\sim f_U}\mathbb{E}_{X\sim P_0^{*,1}}\mathcal{W}_*\left(\rho_X^{*,1}, \rho^{*,2}\right),
\end{aligned}$$

where

$$\begin{aligned}
\mathrm{d}\rho_x^{*,1}(s) &= \frac{C^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\mathbf{1}_{\mathbb{X}\setminus\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s + C\frac{e^{-Cs}}{s}\mathbf{1}_{\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s,\\
\mathrm{d}\rho^{*,2}(s) &= (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s,\\
P_0^{*,1}(\cdot) &= \frac{\sigma U^\sigma}{C}P_0(\cdot) + \frac{1}{C}\sum_{i=1}^{k}(n_i-\sigma)\delta_{x_i^*}(\cdot),\\
P_0^{*,2}(\cdot) &= \frac{\alpha}{\alpha+n}P_0(\cdot) + \frac{1}{\alpha+n}\sum_{i=1}^{n}\delta_{x_i}(\cdot).
\end{aligned}$$

The first term is

$$\mathbb{E}_{U\sim f_U}\mathcal{W}_{d_{\mathbb{X},t}}\left(\frac{\sigma U^\sigma}{C}P_0+\frac{1}{C}\sum_{i=1}^{k_n}(n_i-\sigma)\delta_{x_i^*},\frac{\alpha}{\alpha+n}P_0+\frac{1}{\alpha+n}\sum_{i=1}^{n}\delta_{x_i}\right)=\mathbb{E}_{U\sim f_U}\mathcal{A}^U.$$

The second term is

$$\begin{aligned}\mathbb{E}_{U\sim f_U}\mathbb{E}_{X\sim P_0^{*,1}}\mathcal{W}_*\big(\rho_X^{*,1},\rho^{*,2}\big)&=\mathbb{E}_{U\sim f_U}\frac{\sigma U^\sigma}{C}\mathbb{E}_{X\sim P_0}\mathcal{W}_*\big(\rho_X^{*,1},\rho^{*,2}\big)+\mathbb{E}_{U\sim f_U}\frac{1}{C}\sum_{i=1}^{k_n}(n_i-\sigma)\mathcal{W}_*\Big(\rho_{x_i^*}^{*,1},\rho^{*,2}\Big)\\&=\mathbb{E}_{U\sim f_U}\frac{\sigma U^\sigma}{C}\mathcal{W}_*\left(\frac{C^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\,\mathrm{d}s,(\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\right)\\&\quad+\mathbb{E}_{U\sim f_U}\frac{n-k\sigma}{C}\mathcal{W}_*\left(\frac{Ce^{-Cs}}{s}\,\mathrm{d}s,(\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\right)\\&=\mathbb{E}_{U\sim f_U}J_1^U+\mathbb{E}_{U\sim f_U}J_2^U.\end{aligned}$$

□

Theorem 3.4. *Let $\tilde{\mu}^1$ and $\tilde{\mu}^2$ be as in Proposition 3.3 and let $(x_n)_{n\geq 1}$ be any sequence in $\mathbb{X}$. Then, as $n\to\infty$,*

$$\mathcal{W}_{\mathrm{AD}}\big(\mathcal{L}\left(\tilde{\nu}_S^{*,1}\right),\mathcal{L}\left(\tilde{\nu}_S^{*,2}\right)\big)\asymp\frac{k_n}{n}.$$

In particular, $\mathcal{W}_{\mathrm{BL}}\big(\mathcal{L}\left(\tilde{\mu}_S^{*,1}\right),\mathcal{L}\left(\tilde{\mu}_S^{*,2}\right)\big)\lesssim\frac{k_n}{n}$.

Proof. First, note that the $\mathcal{W}_{\mathrm{BL}}$ bound follows from Theorem 6 of Catalano and Lavenant (2023).

From now on, we attach the subscript n to k, U, and C, writing k_n, U_n, and C_n. Recall that U_n has density function $f_U(u)=\frac{\sigma}{\Gamma\left(k_n+\frac{\theta}{\sigma}\right)}u^{\theta+k_n\sigma-1}e^{-u^\sigma}\mathbf{1}_{(0,\infty)}(u)$. By the change of variables $y=u^\sigma$, the random variable $U_n^\sigma\sim\mathrm{Gamma}\left(k_n+\frac{\theta}{\sigma},1\right)$.

In view of Proposition 3.3, we study the behaviour of each term in $\mathbb{E}_U\left(J_1^{U_n}+J_2^{U_n}+\mathcal{A}^{U_n}\right)$.

- First term: $\mathbb{E}_{U_n}J_1^{U_n}$.

Recall that

$$J_1^{U_n}=\frac{\sigma U_n^\sigma}{C_n}\mathcal{W}_*\left(\frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_ns}}{s^{1+\sigma}}\mathrm{d}s,(\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\mathrm{d}s\right)=\frac{\sigma U_n^\sigma}{C_n}W_n,$$

where $C_n=\sigma U_n^\sigma+n-k_n\sigma$ and

$$W_n=\mathcal{W}_*\left(\frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_ns}}{s^{1+\sigma}}\mathrm{d}s,(\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\mathrm{d}s\right).$$

We first establish an upper bound for the expected value.

The extended Wasserstein distance satisfies

$$
\begin{aligned}
W_n &= \int_0^\infty \left| \int_s^\infty \frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)} \frac{e^{-C_n t}}{t^{1+\sigma}} \mathrm{d}t - \int_s^\infty (\alpha+n) \frac{e^{-(\alpha+n)t}}{t} \mathrm{d}t \right| \mathrm{d}s \\
&\leq \int_0^\infty \int_s^\infty \frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)} \frac{e^{-C_n t}}{t^{1+\sigma}} \mathrm{d}t\, \mathrm{d}s + \int_0^\infty \int_s^\infty (\alpha+n) \frac{e^{-(\alpha+n)t}}{t} \mathrm{d}t\, \mathrm{d}s \\
&\leq \int_0^\infty s \frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)} \frac{e^{-C_n s}}{s^{1+\sigma}} \mathrm{d}s + \int_0^\infty s(\alpha+n) \frac{e^{-(\alpha+n)s}}{s} \mathrm{d}s,
\end{aligned}
$$

where the last inequality follows from Fubini's theorem. Since both integrals equal 1, we conclude that $W_n \leq 2$.

Note also that

$$
\frac{\sigma U_n^\sigma}{C_n} = \frac{\sigma U_n^\sigma}{\sigma U_n^\sigma + n - k_n \sigma} \leq \frac{\sigma}{1-\sigma} \frac{U_n^\sigma}{n}.
$$

Taking expectations, we obtain

$$
\mathbb{E} \frac{\sigma U_n^\sigma}{C_n} \leq \frac{\sigma}{1-\sigma} \frac{k_n + \frac{\theta}{\sigma}}{n}.
$$

Therefore,

$$
\mathbb{E} J_1^{U_n} \leq 2 \frac{\sigma}{1-\sigma} \frac{k_n + \frac{\theta}{\sigma}}{n},
$$

implying that $\mathbb{E} J_1^{U_n} \lesssim \frac{k_n}{n}$.

We now turn to the lower bound. To this end, we distinguish two cases:

$$
\sup k_n < \infty \quad \text{and} \quad \sup k_n = \infty.
$$

- Case: $\sup k_n < \infty$

Since k_n is a nondecreasing sequence of integers, there exists $N \geq 1$ such that $k_n = k$ for all $n \geq N$, where $k = \sup k_n$. This implies that for all $n \geq N$,

$$
U_n^\sigma \sim \text{Gamma}\left(k + \frac{\theta}{\sigma}, 1\right).
$$

Let $p \in (0,1)$ and $t_1 < t_2$ be such that for all $n \geq N$,

$$
\mathbb{P}\left(t_1 < U_n^\sigma < t_2\right) = p.
$$

Denote $A_n = \{t_1 < U_n^\sigma < t_2\}$. Then on A_n,

$$
\frac{\sigma U_n^\sigma}{C_n} = \frac{\sigma U_n^\sigma}{\sigma U_n^\sigma + n - k_n \sigma} \geq \frac{t_1 \sigma}{t_2 \sigma + n - k_n \sigma}.
$$

We now obtain a positive lower bound for W_n on the set A_n.

$$\begin{aligned}
&\mathcal{W}_*\left(\frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_n s}}{s^{1+\sigma}}\mathrm{d}s, (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\mathrm{d}s\right) \\
&\qquad = \int_0^\infty \left| \int_s^\infty \frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_n t}}{t^{1+\sigma}}\mathrm{d}t - \int_s^\infty (\alpha+n)\frac{e^{-(\alpha+n)t}}{t}\mathrm{d}t \right| \mathrm{d}s \\
&\qquad = \int_0^\infty \left| \int_{C_n s}^\infty \frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-y}}{y^{1+\sigma}} C_n^{1+\sigma}\frac{1}{C_n}\mathrm{d}y - \int_{(\alpha+n)s}^\infty (\alpha+n)\frac{e^{-y}}{y}\mathrm{d}y \right| \mathrm{d}s \\
&\qquad = \int_0^\infty \left| \frac{C_n}{\Gamma(1-\sigma)}\Gamma\left(-\sigma, C_n s\right) - (\alpha+n)\Gamma(0,(\alpha+n)s) \right| \mathrm{d}s,
\end{aligned}$$

where $\Gamma(z,t) = \int_t^\infty x^{z-1}e^{-x}\,\mathrm{d}x$ denotes the incomplete gamma function.

By the change of variables $t = C_n s$, the last expression reduces to

$$\int_0^\infty \left| \frac{\Gamma(-\sigma,t)}{\Gamma(1-\sigma)} - \frac{\alpha+n}{C_n}\Gamma\left(0, \frac{\alpha+n}{C_n}t\right) \right| \mathrm{d}t.$$

Note that on A_n,

$$\frac{n}{t_2\sigma+n} \leq \frac{\alpha+n}{\sigma U_n^\sigma + n - k_n\sigma} \leq \frac{\alpha+n}{t_1\sigma + n(1-\sigma)},$$

implying that there exists $N' \geq N$ such that for all $n \geq N'$, on A_n, we have $\frac{\alpha+n}{C_n} \in \left[\frac{1}{2}, c^*\right]$ for some $c^* \in \left(1, \frac{1}{1-\sigma}\right)$. Define the function $F : (0,\infty) \to \mathbb{R}$,

$$F(r) = \int_0^\infty \left| \frac{\Gamma(-\sigma,t)}{\Gamma(1-\sigma)} - r\Gamma\left(0, rt\right) \right| \mathrm{d}t.$$

The function F is continuous. We wish to show that F is bounded below by a positive constant on $\left[\frac{1}{2}, c^*\right]$, which will allow us to deduce that W_n is bounded below on A_n.

Suppose that $F(r) = 0$ for some $r \in \left(\frac{1}{2}, c^*\right)$. Since the integrand is continuous, it follows that for every $t \in (0,\infty)$,

$$\left| \frac{\Gamma(-\sigma,t)}{\Gamma(1-\sigma)} - r\Gamma\left(0, rt\right) \right| = 0.$$

Hence,

$$\frac{\Gamma(-\sigma,t)}{\Gamma(1-\sigma)} = r\Gamma(0,rt).$$

Differentiating with respect to t, we obtain, for all $t > 0$,

$$-\frac{t^{-(1+\sigma)}e^{-t}}{\Gamma(1-\sigma)} = -r^2(rt)^{-1}e^{-rt},$$

that is,

$$t^{-\sigma}e^{t(r-1)} = r\Gamma(1-\sigma).$$

Since the left-hand side is not constant while the right-hand side is, we obtain a contradiction. Therefore, $F(r) > 0$ for all $r \in \left[\frac{1}{2}, c^*\right]$. Since F is continuous, it attains its minimum on this compact interval; denote this minimum by $w > 0$. Consequently, since $\frac{\alpha+n}{C_n} \in \left[\frac{1}{2}, c^*\right]$ on A_n, we have $W_n \geq w$ on A_n.

Combining the above,

$$\mathbb{E}J_1^{U_n} = \mathbb{E}\frac{\sigma U_n^\sigma W_n}{C_n} \geq \mathbb{E}\frac{\sigma U_n^\sigma W_n}{C_n}\mathbf{1}_{A_n} \geq \frac{t_1\sigma}{t_2\sigma + n - k_n\sigma}\, w\, p$$

for $n \geq N'$. Together with the upper bound, we deduce that $\mathbb{E}J_1^{U_n} \asymp \frac{1}{n}$.

- Case: $\sup k_n = \infty$

Our strategy is to show that $\frac{n}{k_n}J_1^{U_n}$ converges in probability to a positive constant and is uniformly integrable. This will yield L^1 convergence and allow us to bound the expectation asymptotically.

We begin by establishing the L^1 convergence of $\frac{U_n^\sigma}{k_n}$ to 1.

$$\mathbb{E}\left|\frac{U_n^\sigma}{k_n} - 1\right|^2 = \mathbb{E}\left|\frac{U_n^\sigma - k_n}{k_n}\right|^2 = \mathbb{E}\left|\frac{U_n^\sigma - \mathbb{E}U_n^\sigma + \frac{\theta}{\sigma}}{k_n}\right|^2 = \frac{\operatorname{Var}(U_n^\sigma)}{k_n^2} + \left(\frac{\frac{\theta}{\sigma}}{k_n}\right)^2.$$

Since $\operatorname{Var}(U_n^\sigma) = k_n + \frac{\theta}{\sigma}$, we get

$$\lim_{n\to\infty} \mathbb{E}\left|\frac{U_n^\sigma}{k_n} - 1\right|^2 = 0.$$

Since L^2 convergence implies L^1 convergence, the desired result follows.

We now show that $\frac{n}{C_n}$ converges in probability to a positive value.

$$\frac{n}{C_n} = \frac{n}{\sigma U_n^\sigma + n - k_n\sigma} = \frac{1}{\frac{\sigma U_n^\sigma}{n} + 1 - \frac{k_n\sigma}{n}}.$$

Note that if $k_n \asymp n$, then by the above, $\frac{U_n^\sigma}{k_n} \xrightarrow{P} 1$, so $\frac{\sigma U_n^\sigma}{n} - \frac{k_n\sigma}{n} \xrightarrow{P} 0$; otherwise both $\frac{U_n^\sigma}{n}$ and $\frac{k_n\sigma}{n}$ converge to 0. In either case, the denominator converges in probability to 1, so

$$\frac{n}{C_n} \xrightarrow{P} 1.$$

We now establish the convergence in probability of W_n. As shown above,

$$W_n = \int_0^\infty \left| \frac{\Gamma(-\sigma, t)}{\Gamma(1-\sigma)} - \frac{\alpha+n}{C_n}\Gamma\left(0, \frac{\alpha+n}{C_n}t\right) \right| \mathrm{d}t.$$

Since $\frac{n}{C_n}$ converges in probability, so does $\frac{\alpha+n}{C_n}$. By the continuous mapping theorem, the last expression converges in probability to

$$\int_0^\infty \left| \frac{\Gamma(-\sigma, t)}{\Gamma(1-\sigma)} - \Gamma(0,t) \right| \mathrm{d}t > 0.$$

We have thus shown that $\frac{n}{k_n} J_1^{U_n}$ converges in probability to some positive constant c. We now establish uniform integrability. Using $W_n \leq 2$ and $C_n \geq n - k_n\sigma$,

$$\frac{n}{k_n}\frac{\sigma U_n^\sigma}{C_n} W_n \leq 2\frac{n\sigma U_n^\sigma}{k_n(n-k_n\sigma)} \leq \frac{n\sigma U_n^\sigma}{k_n n(1-\sigma)} = \frac{\sigma}{1-\sigma}\frac{U_n^\sigma}{k_n}.$$

Since the last term converges in L^1, we obtain uniform integrability.

Therefore,

$$\mathbb{E}\frac{n}{k_n} J_1^{U_n} \to c,$$

from which we deduce that $\mathbb{E}J_1^{U_n} \asymp \frac{k_n}{n}$.

- Second term: $\mathbb{E}_{U_n} J_2^{U_n}$.

Note that

$$\frac{n - k_n\sigma}{C_n} = \frac{n-k_n\sigma}{\sigma U_n^\sigma + n - k_n\sigma} \leq \frac{n-k_n\sigma}{n-k_n\sigma} = 1.$$

It therefore suffices to bound the extended Wasserstein distance.

$$\begin{aligned}
\mathcal{W}_*&\left(C_n \frac{e^{-C_n s}}{s}\mathrm{d}s, (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\mathrm{d}s \right) \\
&= \int_0^\infty \left| \int_s^\infty C_n \frac{e^{-C_n t}}{t}\mathrm{d}t - \int_s^\infty (\alpha+n)\frac{e^{-(\alpha+n)t}}{t}\mathrm{d}t \right| \mathrm{d}s \\
&= \int_0^\infty \left| \int_{C_n s}^\infty C_n \frac{e^{-y}}{y}\mathrm{d}y - \int_{(\alpha+n)s}^\infty (\alpha+n)\frac{e^{-y}}{y}\mathrm{d}y \right| \mathrm{d}s \\
&= \int_0^\infty \left| C_n \Gamma\left(0, C_n s\right) - (\alpha+n)\Gamma(0, (\alpha+n)s) \right| \mathrm{d}s \\
&= J\left(C_n - n, \alpha, n\right),
\end{aligned}$$

where $J\left(\alpha_1, \alpha_2, n\right)$ is defined as

$$J\left(\alpha_1, \alpha_2, n\right) = \int_0^\infty \left| (\alpha_1+n)\Gamma(0, (\alpha_1+n)s) - (\alpha_2+n)\Gamma(0,(\alpha_2+n)s) \right| \mathrm{d}s.$$

By point 2 of Proposition 14 in Catalano and Lavenant (2023), we have that for every $n \geq 1$,

$$J\left(C_n - n, \alpha, n\right) \leq c \max\left\{\log\left(\frac{C_n}{\alpha+n}\right), \log\left(\frac{\alpha+n}{C_n}\right)\right\},$$

where $c = \int_0^\infty |\Gamma(0,t) - e^{-t}| \, \mathrm{d}t$.

Using the fact that $\log x \leq x - 1$, we have that

$$J\left(C_n - n, \alpha, n\right) \leq c \max\left\{\frac{C_n}{\alpha+n} - 1, \frac{\alpha+n}{C_n} - 1\right\}.$$

We bound each term in the maximum separately.

$$\frac{C_n}{\alpha+n} - 1 = \frac{\sigma U_n^\sigma - k_n\sigma - \alpha}{\alpha+n} \leq \sigma \frac{U_n^\sigma}{\alpha+n}.$$

Since

$$\mathbb{E}\frac{\sigma U_n^\sigma}{\alpha+n} = \frac{\sigma\left(k_n + \frac{\theta}{\sigma}\right)}{\alpha+n},$$

we have that

$$\mathbb{E}\left(\frac{C_n}{\alpha+n} - 1\right) \lesssim \frac{k_n}{n}.$$

We now bound the second term in the maximum.

$$\frac{\alpha+n}{C_n} - 1 = \frac{\alpha - \sigma U_n^\sigma + k_n\sigma}{n + \sigma U_n^\sigma - k_n\sigma} \leq \frac{\alpha + k_n\sigma}{n(1-\sigma)}.$$

Therefore,

$$\mathbb{E}\left(\frac{\alpha+n}{C_n} - 1\right) \lesssim \frac{k_n}{n}.$$

- Third term: $\mathcal{A}^{U_n}$.

Finally, we bound

$$\mathcal{A}^{U_n} = \mathcal{W}_{\mathrm{d}_{\mathbb{X},t}}\left(\frac{\sigma U_n^\sigma}{C_n} P_0 + \frac{1}{C_n}\sum_{i=1}^{k_n}(n_i - \sigma)\delta_{x_i^*}, \frac{\alpha}{\alpha+n} P_0 + \frac{1}{\alpha+n}\sum_{i=1}^{n}\delta_{x_i}\right)$$

The proof uses two convexity properties of the Wasserstein distance. The first is that for all $\lambda \in (0,1)$ and $P^1, P^2, Q^1, Q^2 \in \mathcal{P}(\mathbb{X})$,

$$\mathcal{W}_{d_{\mathbb{X},t}}\left(\lambda P^1 + (1-\lambda)Q^1, \lambda P^2 + (1-\lambda)Q^2\right) \leq \lambda \mathcal{W}_{d_{\mathbb{X},t}}(P^1, P^2) + (1-\lambda)\mathcal{W}_{d_{\mathbb{X},t}}(Q^1, Q^2).$$

The second is that for all $\lambda \in (0,1)$ and $P, Q^1, Q^2 \in \mathcal{P}(\mathbb{X})$,

$$\mathcal{W}_{d_{\mathbb{X},t}}(\lambda P + (1-\lambda)Q^1, \lambda P + (1-\lambda)Q^2) = (1-\lambda)\mathcal{W}_{d_{\mathbb{X},t}}(Q^1, Q^2).$$

Since both measures share P_0, we aim to match the coefficient of P_0 in order to apply the second property. Since the relative magnitudes of these coefficients are not known a priori, we consider two cases:

$$\frac{\sigma U_n^\sigma}{C_n} > \frac{\alpha}{\alpha+n} \quad \text{and} \quad \frac{\sigma U_n^\sigma}{C_n} < \frac{\alpha}{\alpha+n}.$$

We decompose the expectation over the sets where these conditions hold. As shown below, on both sets the expectations are asymptotically bounded by

$$\frac{k_n}{n}.$$

Define the set

$$A_n = \left\{ \frac{\sigma U_n^\sigma}{C_n} > \frac{\alpha}{\alpha+n} \right\}.$$

It follows that

$$\mathbb{E}\mathcal{A}^U = \mathbb{E}\mathcal{A}^U \mathbf{1}_{A_n} + \mathbb{E}\mathcal{A}^U \mathbf{1}_{A_n^c}.$$

On A_n, we decompose

$$\frac{\sigma U_n^\sigma}{C_n} P_0 = \frac{\alpha}{\alpha+n} P_0 + \left(\frac{\sigma U_n^\sigma}{C_n} - \frac{\alpha}{\alpha+n} \right) P_0 = \frac{\alpha}{\alpha+n} P_0 + \lambda_1 P_0,$$

where

$$\lambda_1 = \frac{\sigma U_n^\sigma}{C_n} - \frac{\alpha}{\alpha+n}.$$

Note that $\lambda_1 \in (0,1)$. Then

$$\begin{aligned}
&\mathcal{W}_{d_{\mathbb{X}},t}\Big(\frac{\alpha}{\alpha+n} P_0 + \frac{n}{\alpha+n} \left(\frac{\alpha+n}{n} \lambda_1 P_0 + \frac{\alpha+n}{n} \frac{n-k_n\sigma}{C_n} \sum_{i=1}^{k_n} \frac{(n_i-\sigma)}{n-k_n\sigma} \delta_{x_i^*} \right), \\
&\qquad \frac{\alpha}{\alpha+n} P_0 + \frac{n}{\alpha+n} \sum_{i=1}^{n} \frac{1}{n} \delta_{x_i} \Big) \\
&\qquad = \frac{n}{\alpha+n} \mathcal{W}_{d_{\mathbb{X}},t} \left(\frac{\alpha+n}{n} \lambda_1 P_0 + \frac{\alpha+n}{n} \frac{n-k_n\sigma}{C_n} \sum_{i=1}^{k_n} \frac{(n_i-\sigma)}{n-k_n\sigma} \delta_{x_i^*}, \sum_{i=1}^{n} \frac{1}{n} \delta_{x_i} \right) \\
&\qquad = \frac{n}{\alpha+n} \mathcal{W}_{d_{\mathbb{X}},t} \Big(\frac{\alpha+n}{n} \lambda_1 P_0 + \frac{\alpha+n}{n} \frac{n-k_n\sigma}{C_n} \sum_{i=1}^{k_n} \frac{(n_i-\sigma)}{n-k_n\sigma} \delta_{x_i^*}, \\
&\qquad \frac{\alpha+n}{n} \lambda_1 \sum_{i=1}^{n} \frac{1}{n} \delta_{x_i} + \frac{\alpha+n}{n} \frac{n-k_n\sigma}{C_n} \sum_{i=1}^{n} \frac{1}{n} \delta_{x_i} \Big) \\
&\qquad \leq \lambda_1 \mathcal{W}_{d_{\mathbb{X}},t} \left(P_0, \sum_{i=1}^{n} \frac{1}{n} \delta_{x_i} \right) + \lambda_2 \mathcal{W}_{d_{\mathbb{X}},t} \left(\sum_{i=1}^{k_n} \frac{(n_i-\sigma)}{n-k_n\sigma} \delta_{x_i^*}, \sum_{i=1}^{n} \frac{1}{n} \delta_{x_i} \right),
\end{aligned}$$

where

$$\lambda_2 = \frac{n - k_n\sigma}{C_n}.$$

We have shown that

$$\mathbb{E}\mathcal{A}^{U_n}\mathbf{1}_{A_n} \leq \mathbb{E}\left[\mathbf{1}_{A_n}\lambda_1\mathcal{W}_{d_{\mathbb{X},t}}\left(P_0, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right) + \mathbf{1}_{A_n}\lambda_2\mathcal{W}_{d_{\mathbb{X},t}}\left(\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*}, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right)\right].$$

In the first term inside the expectation, since $d_{\mathbb{X},t} \leq 2$ we can bound the Wasserstein distance by 2 and λ_1 by $\frac{\sigma U_n^\sigma}{C_n}$. Hence,

$$\mathbb{E}\left[\mathbf{1}_{A_n}\lambda_1\mathcal{W}_{d_{\mathbb{X},t}}\left(P_0, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right)\right] \leq \mathbb{E}\frac{\sigma U_n^\sigma}{C_n} \leq \frac{\sigma}{1-\sigma}\frac{k_n + \frac{\theta}{\sigma}}{n},$$

so this term is asymptotically bounded by $\frac{k_n}{n}$.

We now bound

$$\mathbb{E}\left[\mathbf{1}_{A_n}\lambda_2\mathcal{W}_{d_{\mathbb{X},t}}\left(\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*}, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right)\right].$$

Since $C_n \geq n - k_n\sigma$, we have $\lambda_2 \leq 1$, and the Wasserstein distance does not depend on U_n. Note also that $\sum_{i=1}^{n}\frac{1}{n}\delta_{x_i} = \sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}$. Since we can decompose

$$\sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*} = \sum_{i=1}^{k_n}\frac{n_i-\sigma}{n}\delta_{x_i^*} + \sum_{i=1}^{k_n}\frac{\sigma}{n}\delta_{x_i^*} = \frac{n-k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*} + \frac{k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{1}{k_n}\delta_{x_i^*},$$

applying the convexity of the Wasserstein distance, we obtain

$$\begin{aligned}
\mathcal{W}_{d_{\mathbb{X},t}}&\left(\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*}, \sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}\right) = \\
&= \mathcal{W}_{d_{\mathbb{X},t}}\left(\frac{n-k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*} + \frac{k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*},\right. \\
&\qquad \left.\frac{n-k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*} + \frac{k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{1}{k_n}\delta_{x_i^*}\right) \\
&= \frac{k_n\sigma}{n}\mathcal{W}_{d_{\mathbb{X},t}}\left(\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*}, \sum_{i=1}^{k_n}\frac{1}{k_n}\delta_{x_i^*}\right) \\
&\leq 2\frac{k_n\sigma}{n}.
\end{aligned}$$

We conclude that $\mathbb{E}\mathcal{A}^{U}\mathbf{1}_{A_n} \lesssim \frac{k_n}{n}$.

Now suppose we are on A_n^c. We decompose

$$\frac{\alpha}{\alpha+n}P_0 = \frac{\sigma U_n^\sigma}{C_n}P_0 + \left(\frac{\alpha}{\alpha+n} - \frac{\sigma U_n^\sigma}{C_n}\right)P_0.$$

Define

$$\lambda_1' = \frac{\alpha}{\alpha+n} - \frac{\sigma U_n^\sigma}{C_n}.$$

Note that, on A_n^c, $\lambda_1' \in (0,1)$. On A_n^c,

$$\begin{aligned}
&\mathcal{W}_{d_{\mathbb{X}},t}\Bigg(\frac{\sigma U_n^\sigma}{C_n}P_0 + \frac{n-k_n\sigma}{C_n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*}, \\
&\qquad\qquad \frac{\sigma U_n^\sigma}{C_n}P_0 + \frac{n-k_n\sigma}{C_n}\left(\frac{C_n}{n-k_n\sigma}\lambda_1' P_0 + \frac{C_n}{n-k_n\sigma}\frac{n}{\alpha+n}\sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}\right)\Bigg) \\
&\qquad = \frac{n-k_n\sigma}{C_n}\mathcal{W}_{d_{\mathbb{X}},t}\left(\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*}, \frac{C_n}{n-k_n\sigma}\lambda_1' P_0 + \frac{C_n}{n-k_n\sigma}\frac{n}{\alpha+n}\sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}\right) \\
&\qquad \le \lambda_1'\mathcal{W}_{d_{\mathbb{X}},t}\left(\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*}, P_0\right) + \frac{n}{\alpha+n}\mathcal{W}_{d_{\mathbb{X}},t}\left(\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*}, \sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}\right).
\end{aligned}$$

Since $\lambda_1' \le \frac{\alpha}{\alpha+n}$, the first term is asymptotically bounded by $\frac{k_n}{n}$. As we have already bounded the second Wasserstein distance, we deduce that $\mathbb{E}\mathcal{A}^U\mathbf{1}_{A_n^c} \lesssim \frac{k_n}{n}$. □

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